



Lebanese University
Faculty of Sciences
Master of Science in Health Administration



Introduction to AI

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What is artificial intelligence?

■ Scientific definition:

Artificial intelligence (AI) is the ability of digital systems to perform tasks that usually require human intelligence, such as: Understanding, learning, planning, decision-making, and interaction with language and image.

■ Academic definition (UNESCO):

Artificial intelligence (AI) is a set of digital systems capable of performing tasks that normally require human intelligence, such as perception, learning, problem-solving, and interaction with the environment.

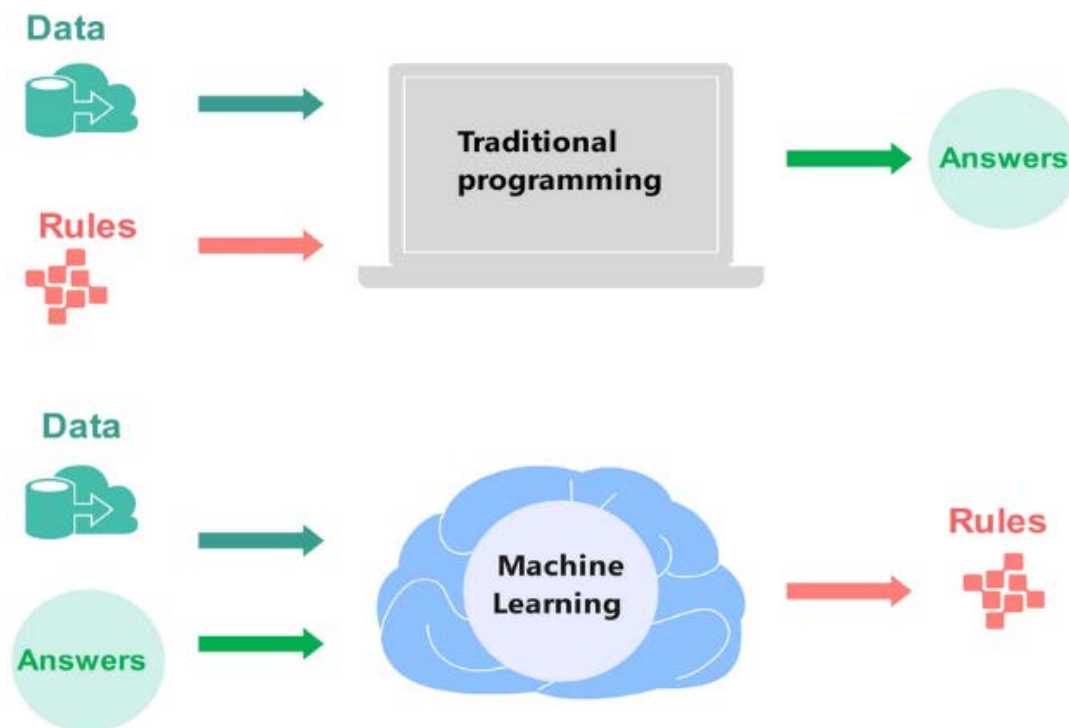


Overview

What is AI?

Intelligence: ability to **extract knowledge** from observations

This knowledge is used to **solve tasks in different contexts and environments** (automation)





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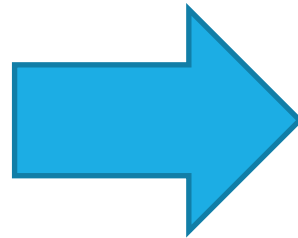
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Old way: Memorize

- Human experts code the machines
- **Goods:** we know what we are doing.
- **Bads:** requires **explicit** solutions (not available for some problems).



Modern way: Generalize

- Let machines teach themselves how to solve a problem (**implicit**).
- **Goods:** universally applicable
- **Bads:** lack of **understandability/robustness**.
- Requires **training**.

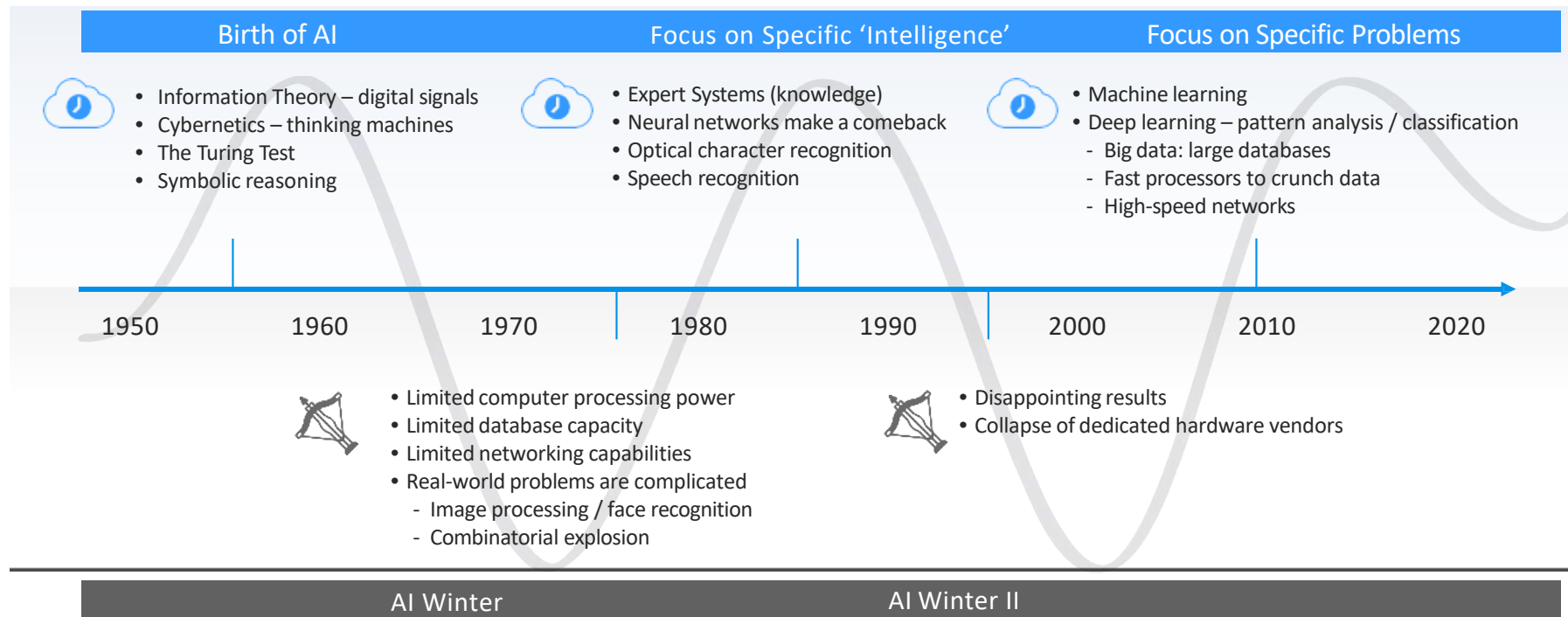
The development of artificial intelligence

■ Time periods

Stage	time period	Key Features
First stage	1950s - 1970s	Early attempts to simulate human thought (Turing test, simple software)
Phase Two	1980s - 1990s	Expert Intelligence Systems, Knowledge Base Processing
Phase Three	2000–2010	The emergence of big data and the internet, machine learning
Phase Four	2010–Present	Deep Learning , Generative AI

AI Timeline

An AI Timeline



The great elders of modern AI (Turing Prize 2018)

Geoffrey Hinton



- Cognitive psychologist and computer scientist,
- Prof. at University of Toronto and works for Google,
- Known for back-propagation and Boltzmann machines.

Yoshua Bengio



- Computer scientist,
- Prof. at Université de Montréal and head of MILA,
- Known for his work on deep learning.

Yann le Cun



- Computer scientist,
- Prof. at New York University then he joins FAIR,
- Known for his work on back-propagation and CNNs.

Where did the revolution in AI come from?

- The use of GPUs for computation.
- The share of huge datasets on Internet.
- Github/Arxiv new ways of sharing research.
- The return of representation learning.



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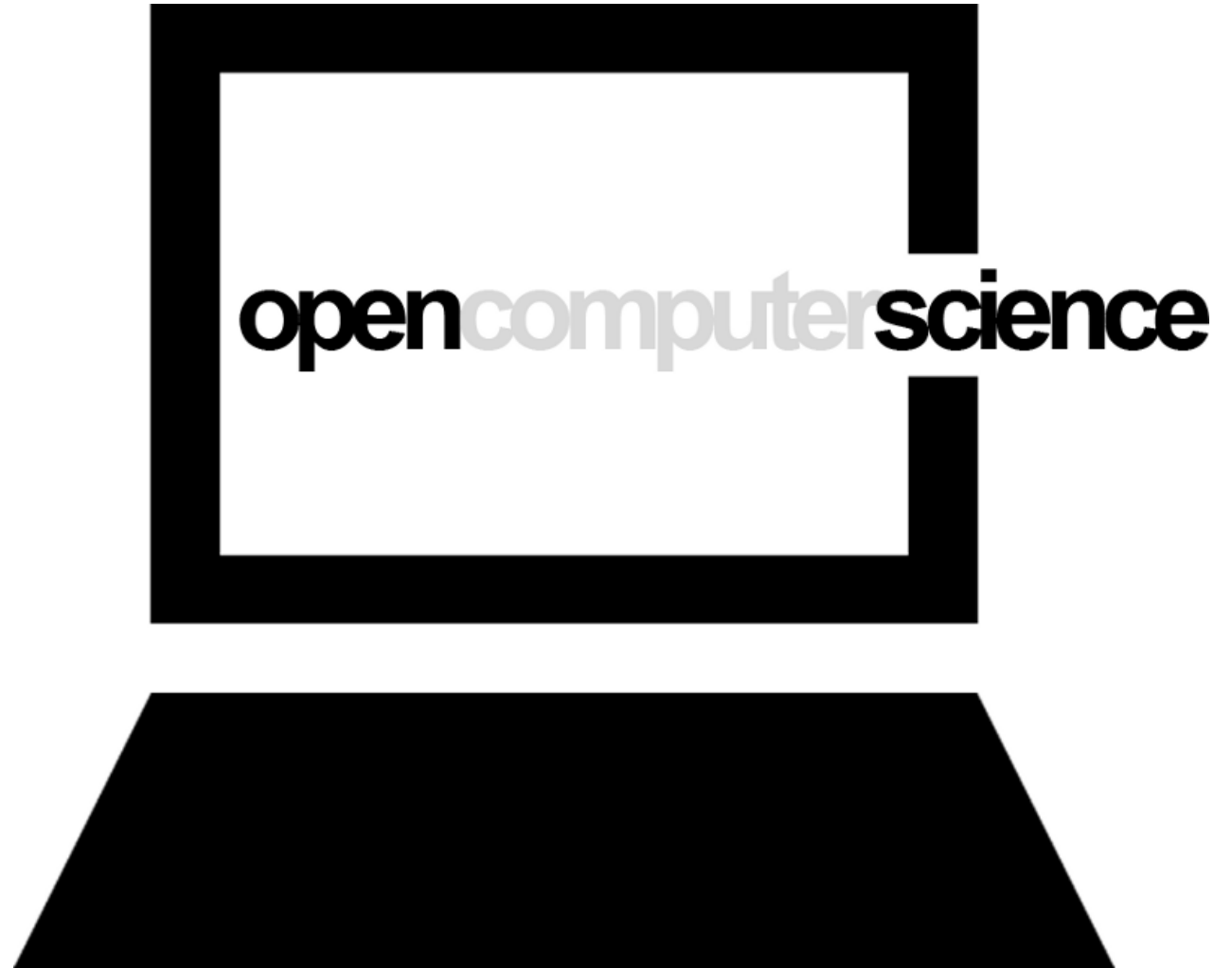


1. NIH Database of 100,000 Chest X-Rays: <https://nihcc.app.box.com/v/ChestXray-NIHCC>
2. The Cancer Imaging Data Set: <https://www.cancerimagingarchive.net/>
3. CT SCAN IMAGES: <https://www.kaggle.com/kmader/siim-medical-images>
4. Chest x-rays: <https://www.kaggle.com/paultimothymooney/chest-xray-pneumonia>



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■ Traditional Pattern Recognition: Fixed/Handcrafted Feature Extractor



■ Mainstream Modern Pattern Recognition: Unsupervised mid-level features



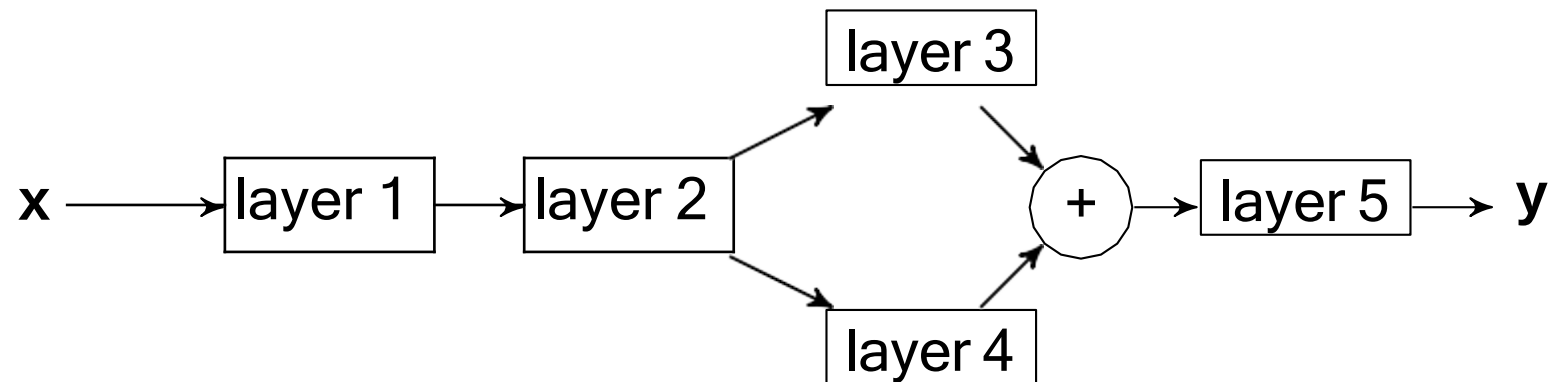
■ Deep Learning: Representations are hierarchical and trained



Where did the revolution in AI come from?

Main idea

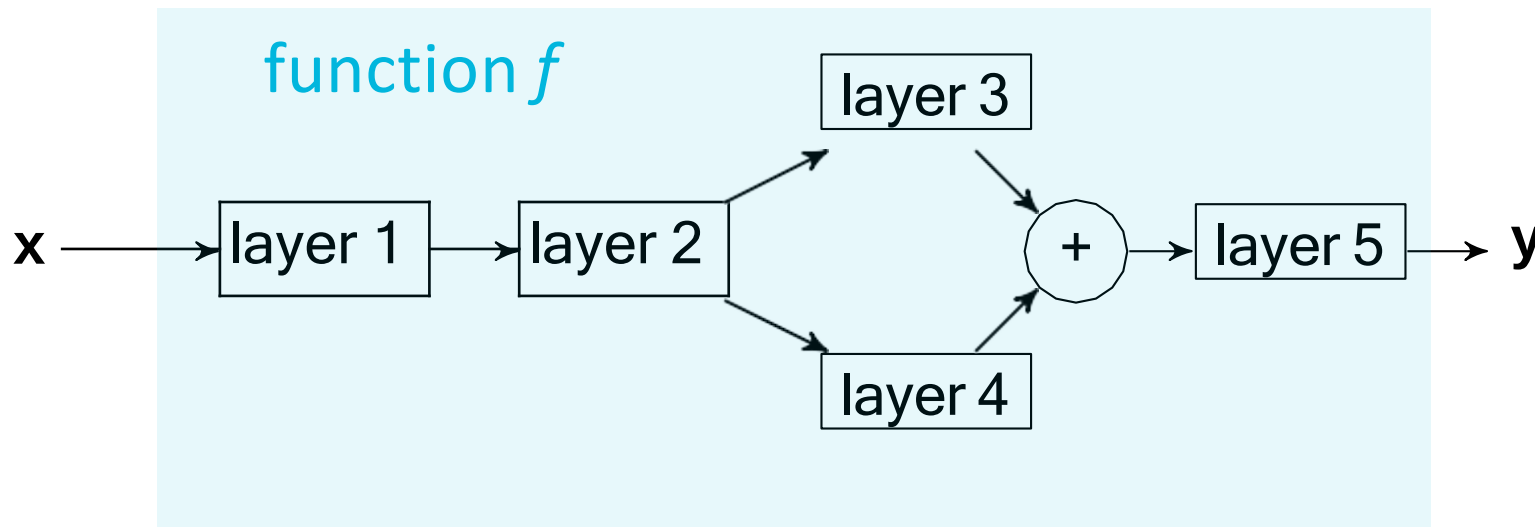
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- **End-to-end learning:** Tune all atomic functions together
- **Training:** Backpropagate throughout the architecture (to compute the gradient of the loss wrt all layers parameters)



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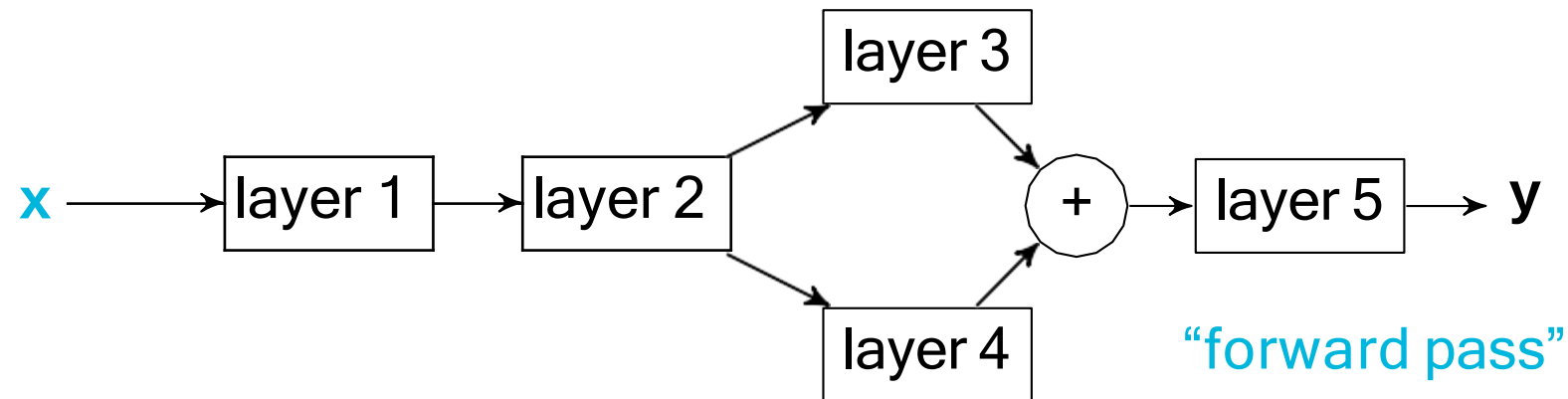
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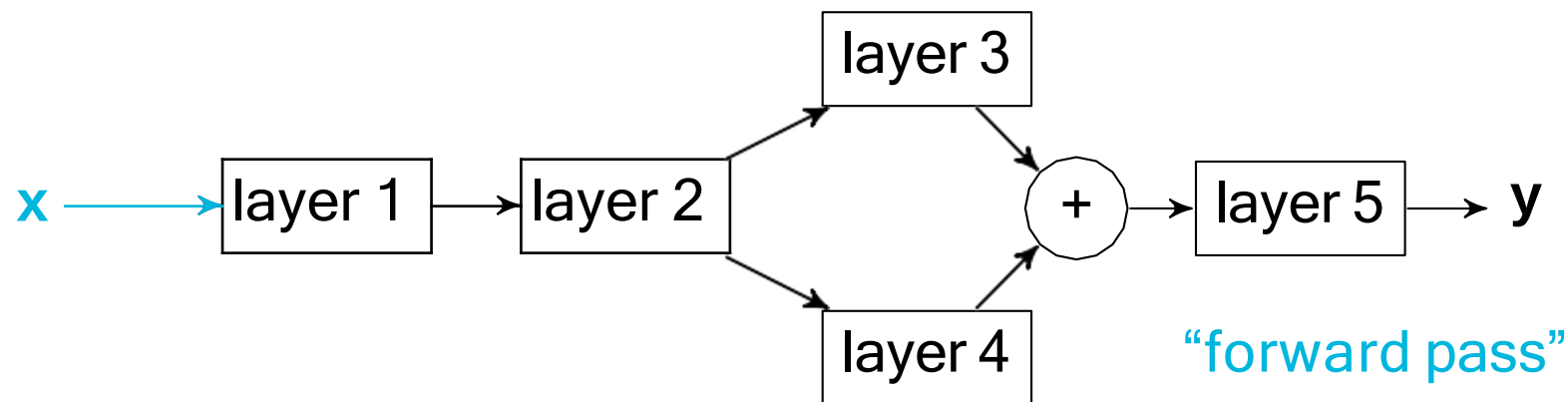
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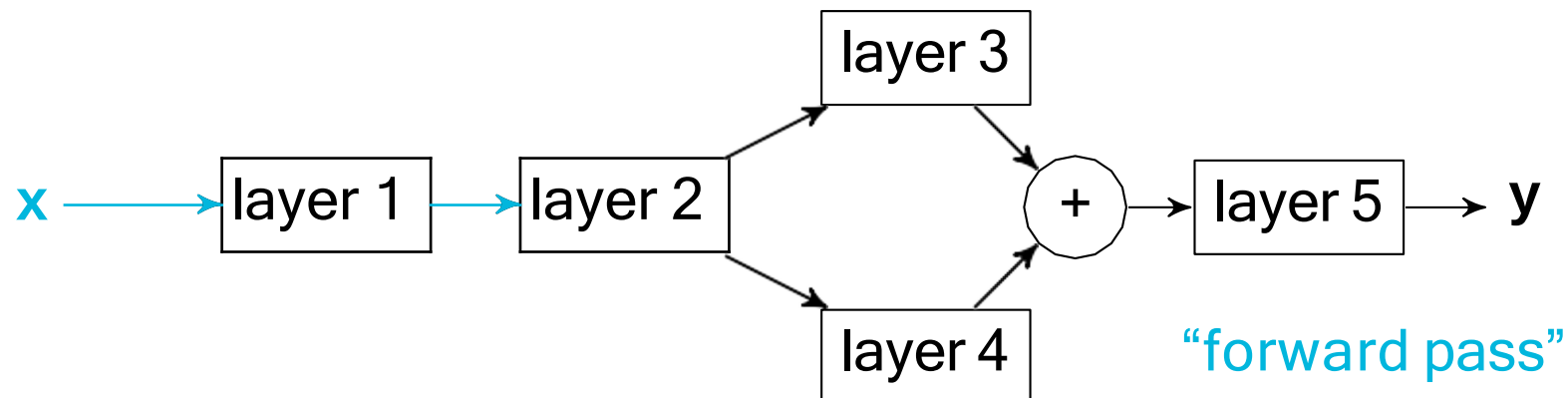
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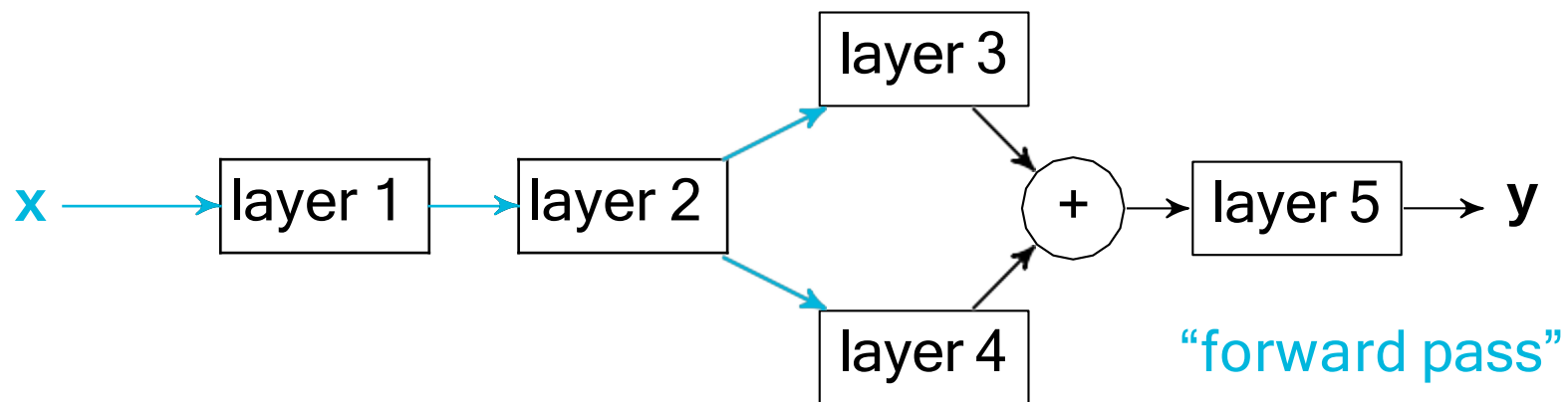
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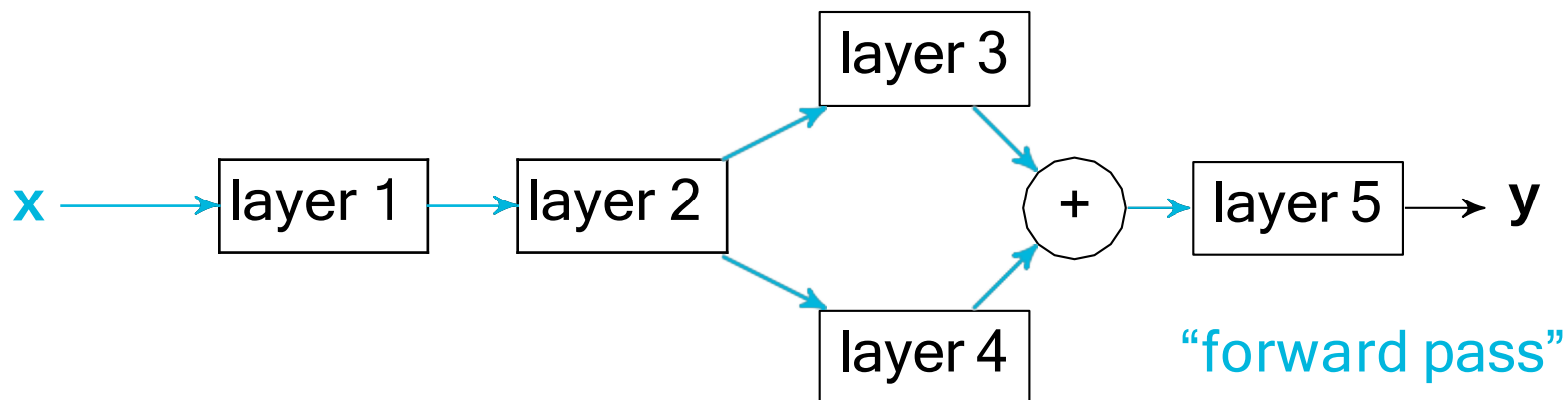
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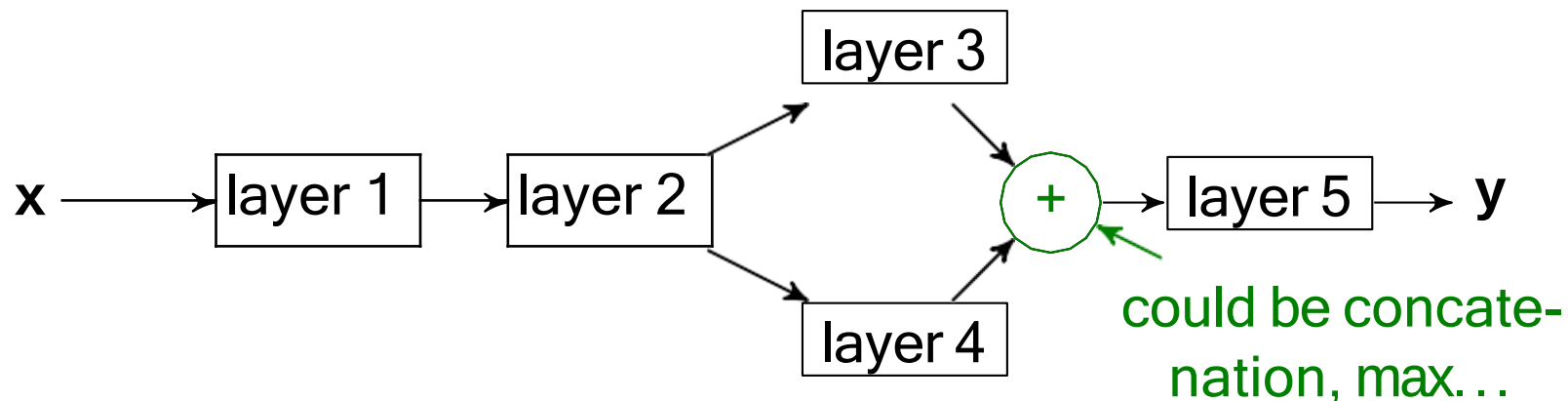
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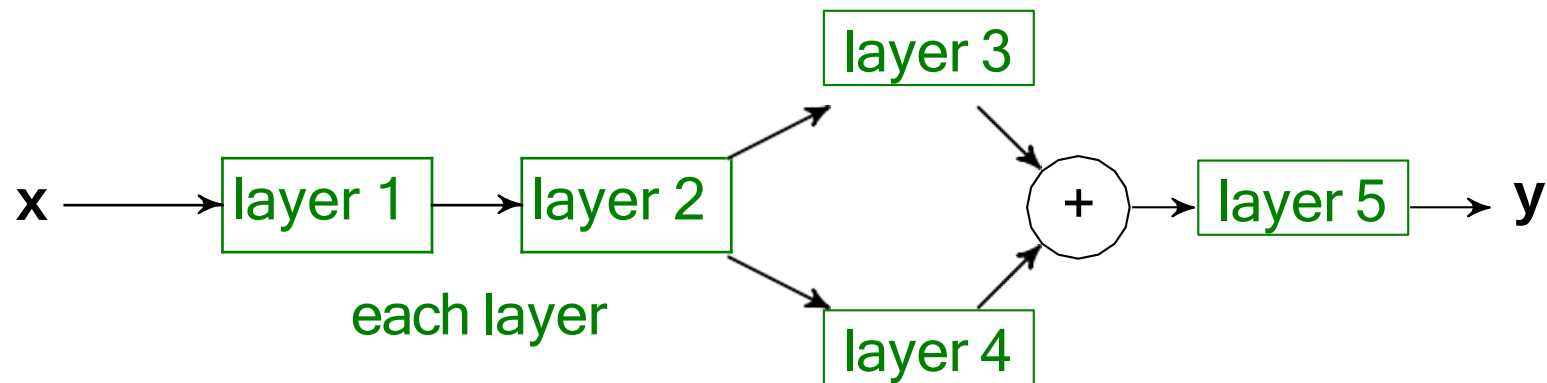
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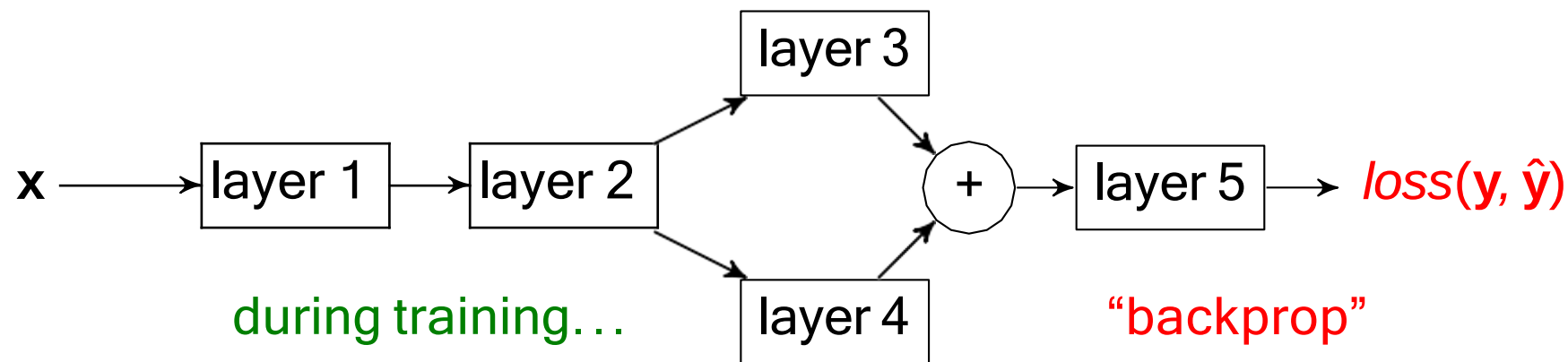
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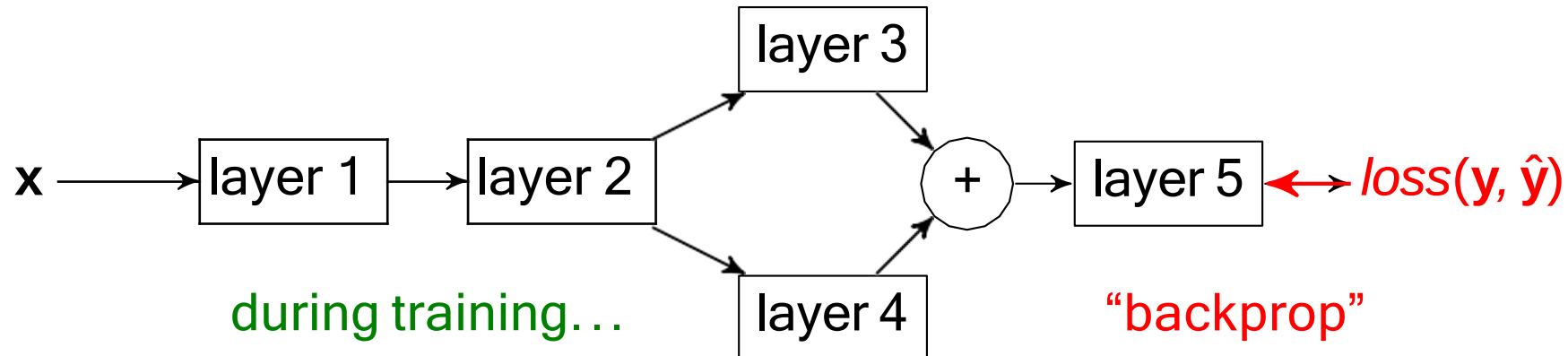
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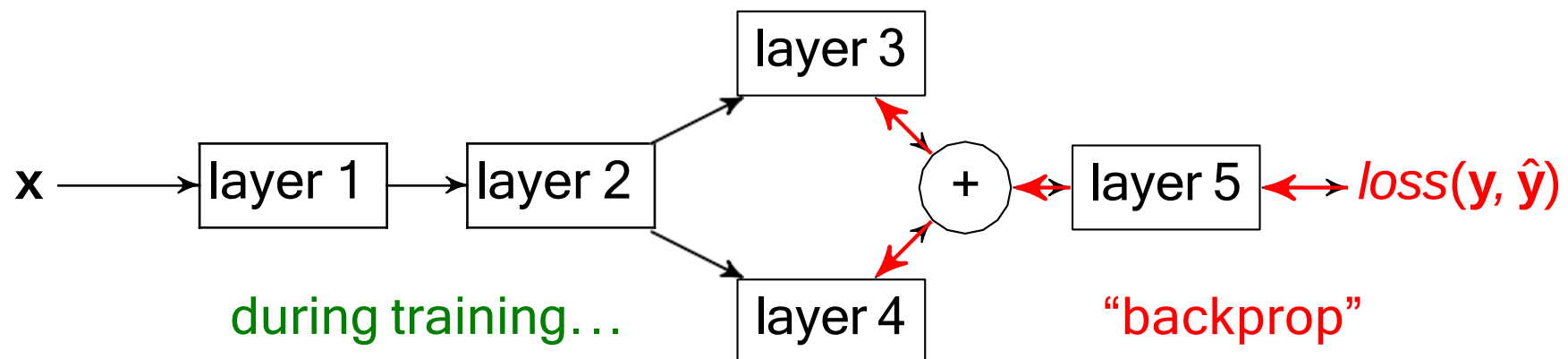
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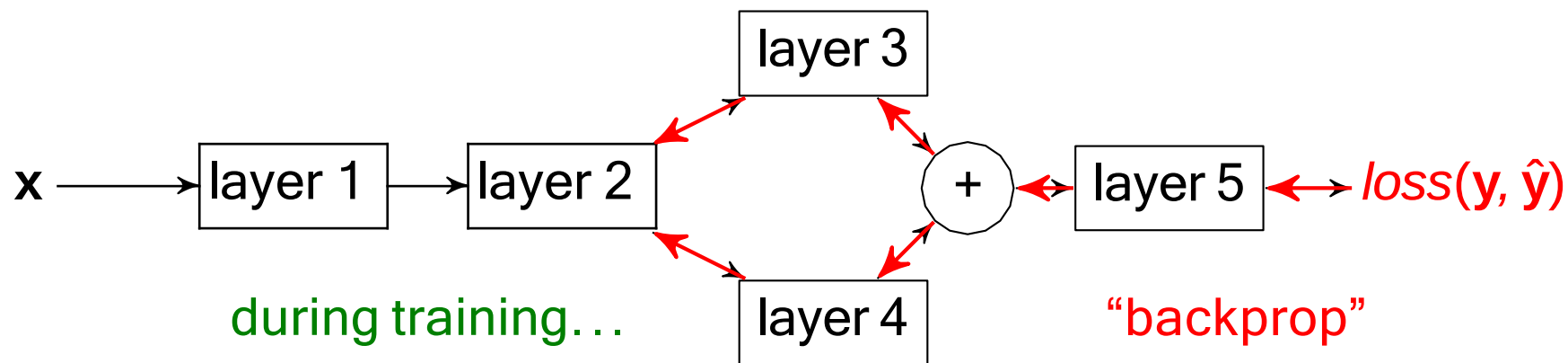
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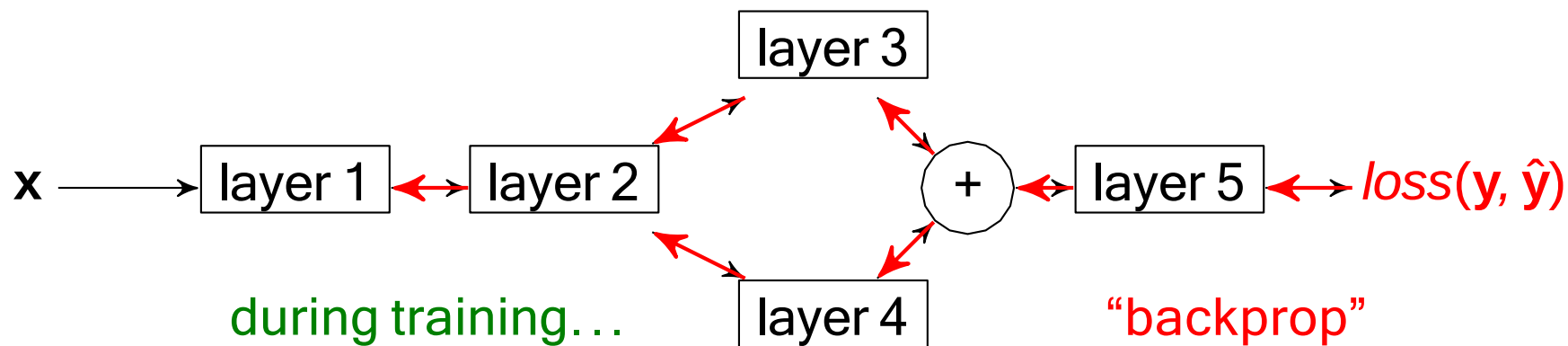
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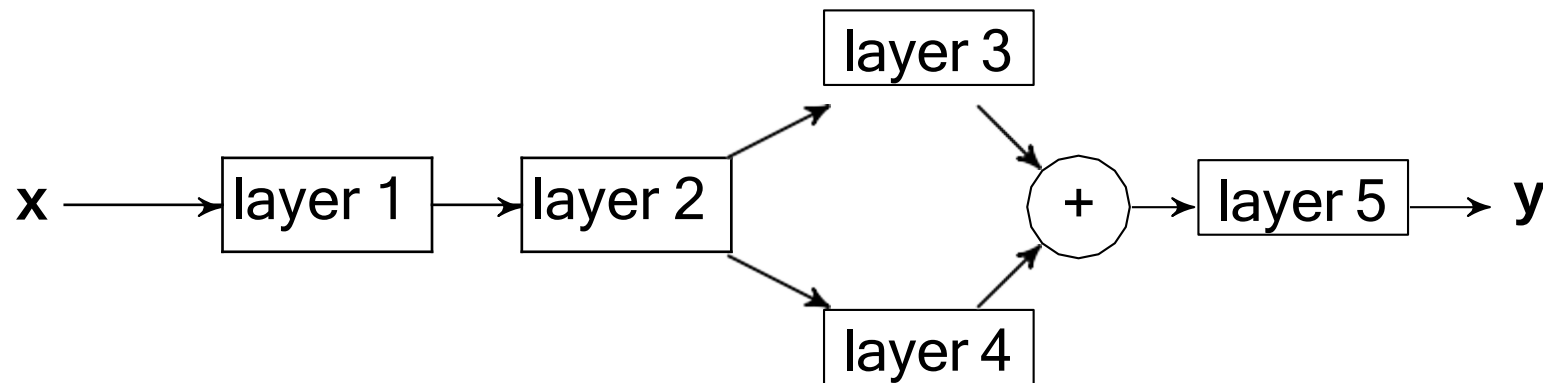
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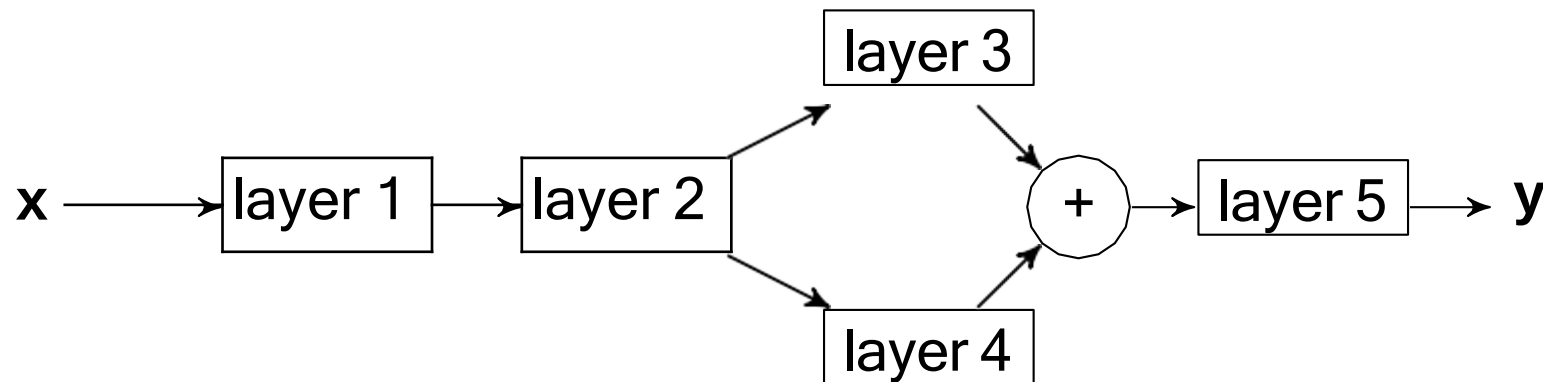


Number of layers, choice of the architecture are **hyperparameters**

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Where did the revolution in AI come from?

Layers

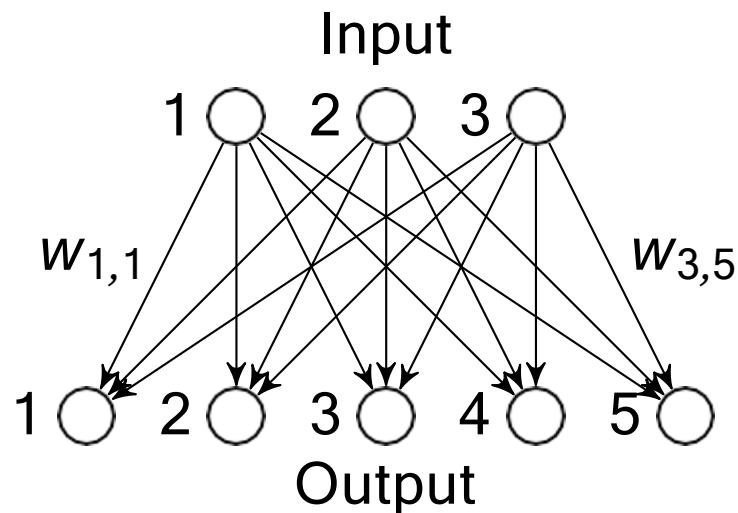
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 - h is a nonlinear parameterwise function (often without parameters),
 - $\mathbf{W}$ is a tensor:
 - Can be agnostic of the structure: **fully-connected layers**,
 - Can be structure-dependent: **convolutional layers**.

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Fully connected layer

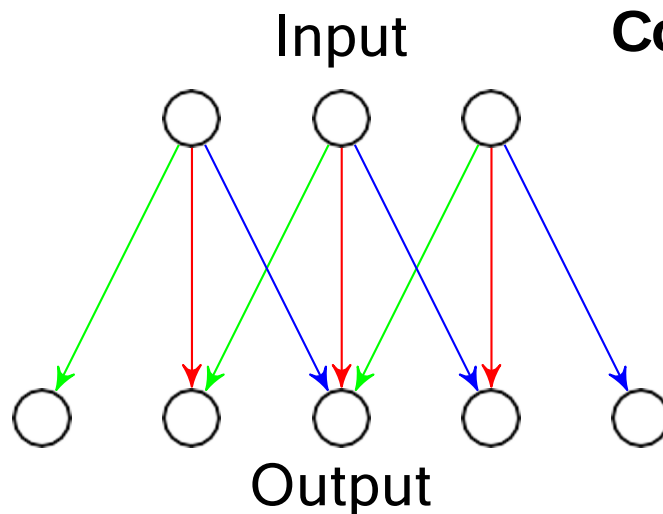


$W_{1,1}$	$W_{1,2}$	$W_{1,3}$	$W_{1,4}$	$W_{1,5}$
$W_{2,1}$	$W_{2,2}$	$W_{2,3}$	$W_{2,4}$	$W_{2,5}$
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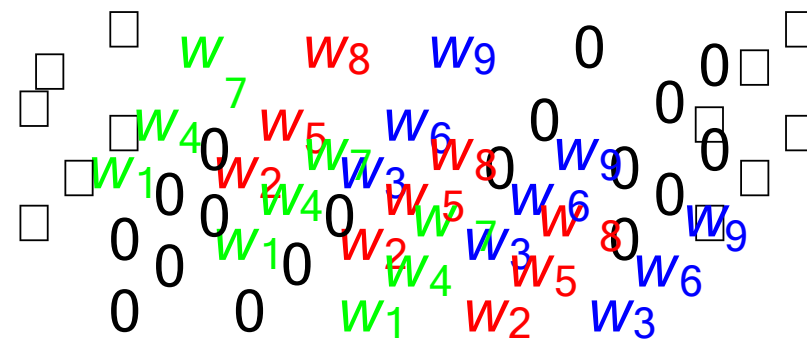
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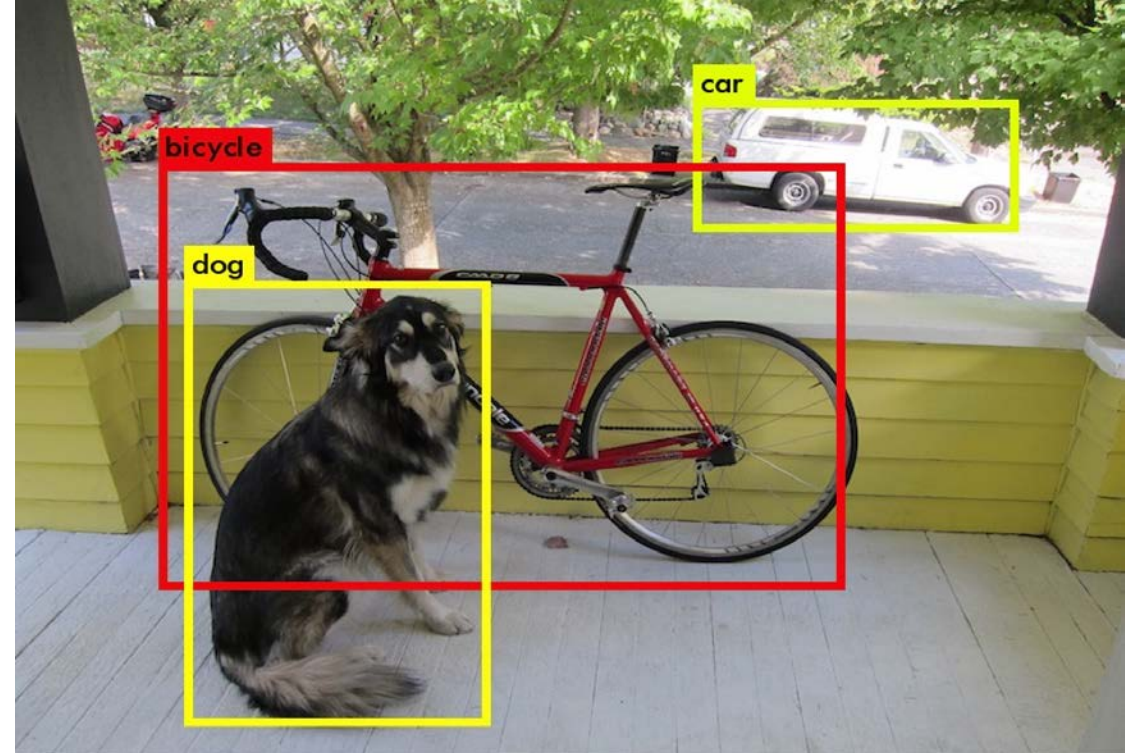
Convolutional layer



Main application domains of AI

Vision

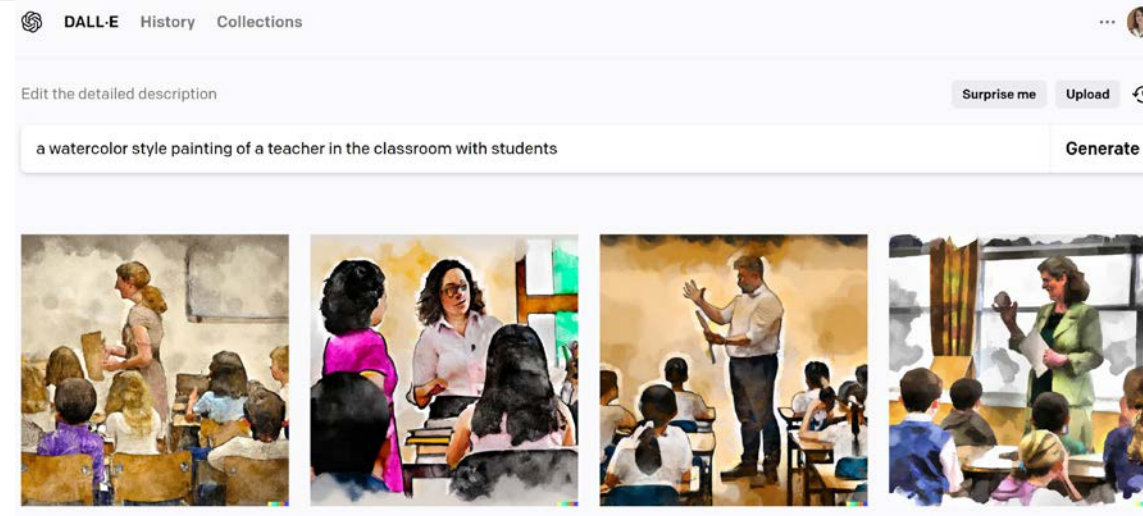
- Object/face recognition,
- Detection,
- Autonomous vehicles,
- Automatic diagnostic,
- Defects identification,
- Video applications...



Main application domains of AI

Recent breakthrough: Image Synthesis

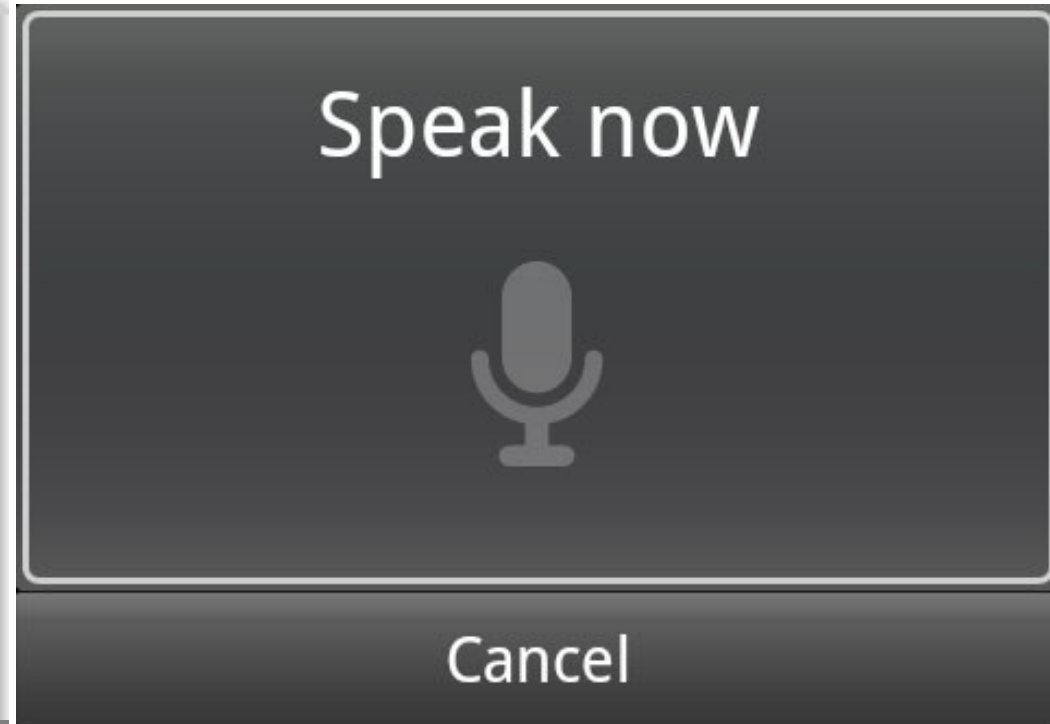
- Stable Diffusion Models: state of the art image generation (<https://arxiv.org/abs/2112.10752>)
- CLIP: Learning visual models from Natural Language Supervision (<https://arxiv.org/abs/2103.00020>)
- DALL-E (<https://labs.openai.com/>), Midjourney (<https://midjourney.gitbook.io/docs/>), Stable Diffusions



Main application domains of AI

Natural Language Processing (NLP)

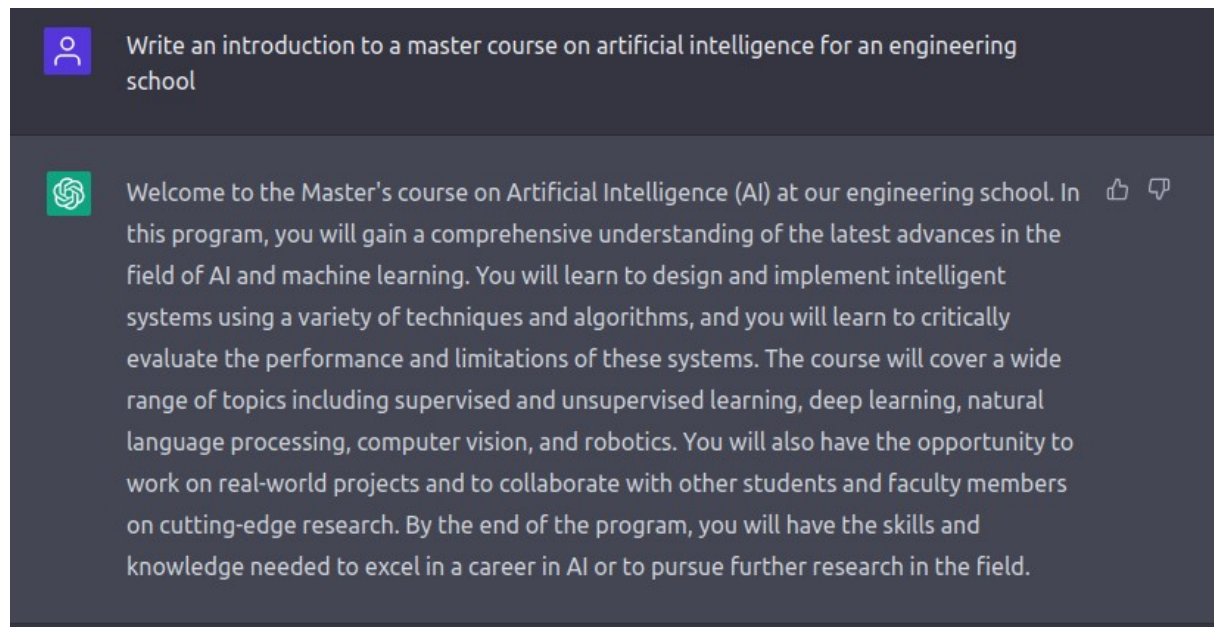
- Automatic assistant,
- Voice-to-text,
- Automatic translation,
- Automatic summarizing,
- Sentiment analysis,
- Text indexing...



Main application domains of AI

Recent breakthrough: Text Synthesis with Large Language Models

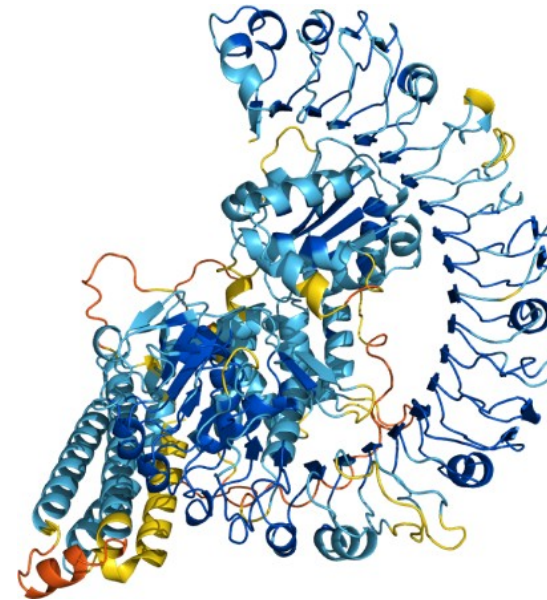
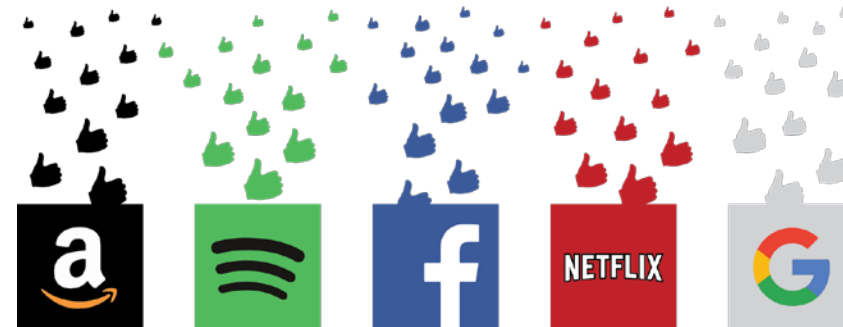
- GPT-3, 175 billion parameters trained without explicit supervision on over 600GB of text <https://arxiv.org/abs/2005.14165>
- ChatGPT <https://openai.com/blog/chatgpt/>



Main application domains of AI

Tons of other domains...

- Medical imaging,
- Decision aid,
- Data mining,
- Visualization,
- Recommender systems,
- Market analysis...

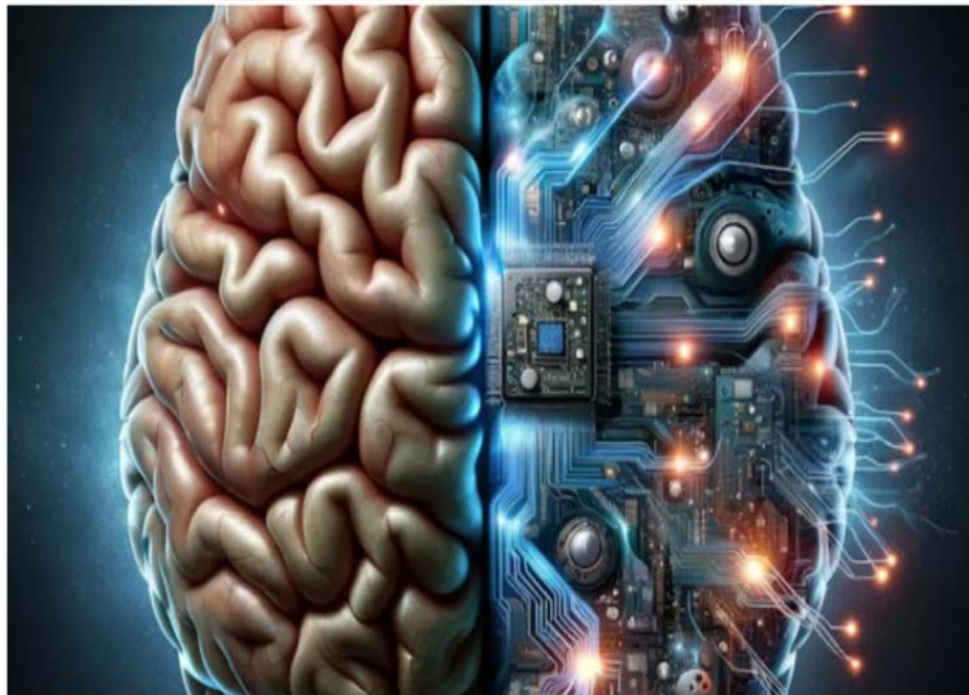


<https://alphafold.ebi.ac.uk/>

How AI Works

■ Basic concept

- Artificial intelligence **does not "think"** like a human, but **it mimics** our thinking pattern by analyzing data and learning from past experiences.

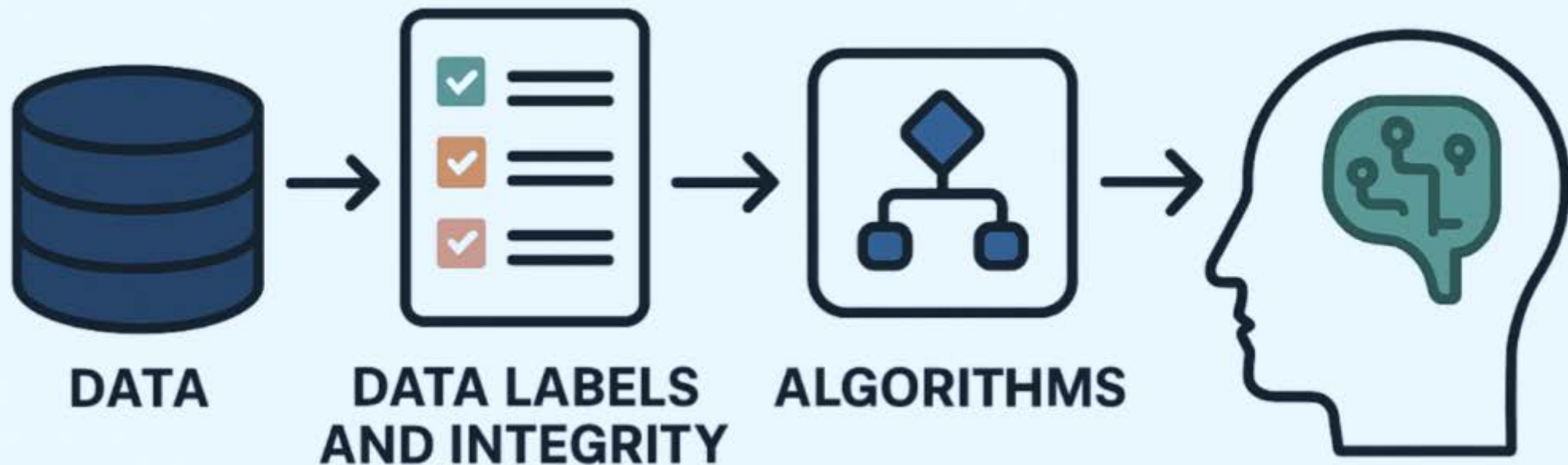


How AI Works

■ Basic concept

- The operating principle of any artificial intelligence system can be summarized in the following equation:

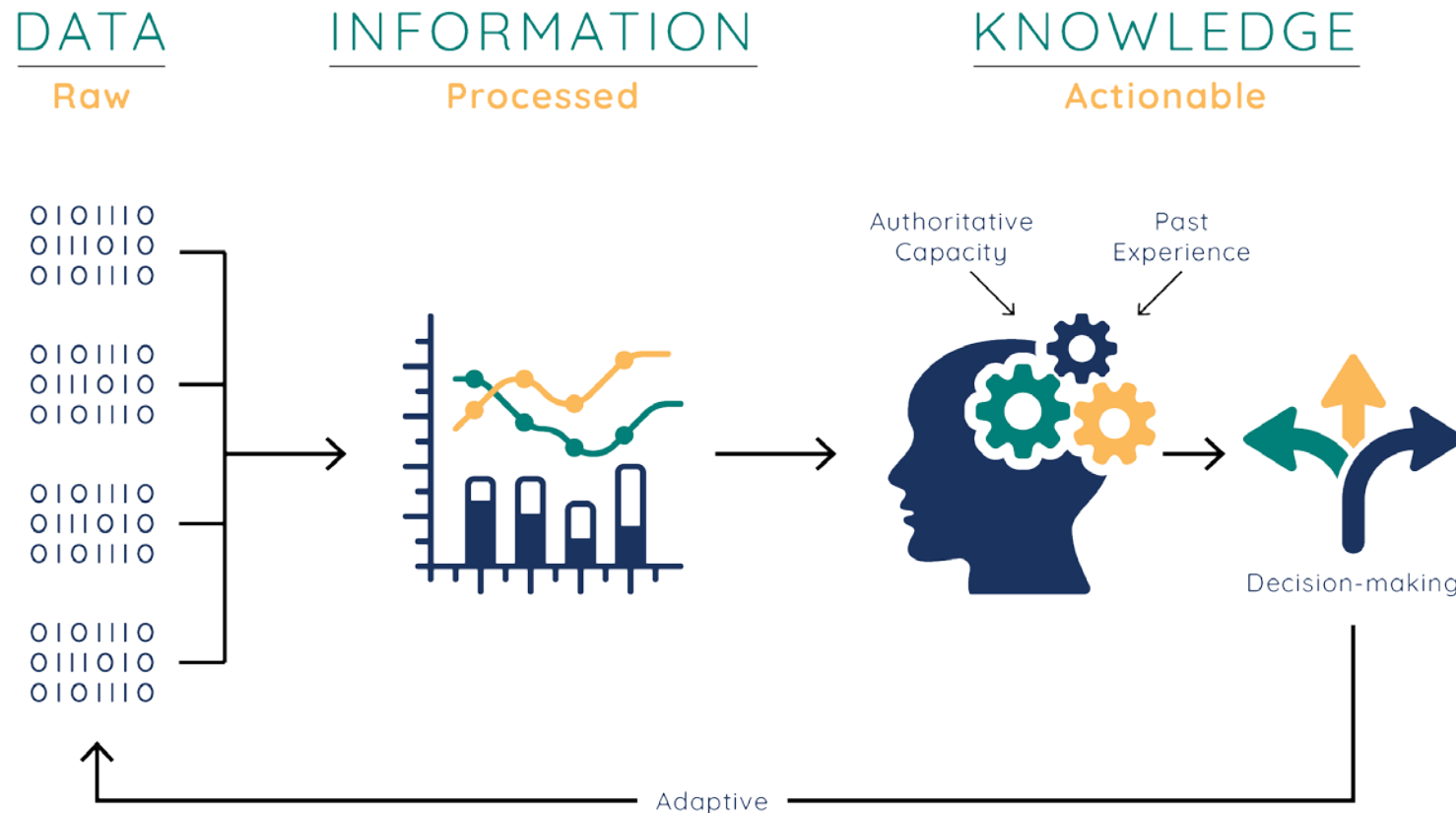
Data + Algorithms + Learning = Intelligence



How AI Works

■ How does artificial intelligence learn?

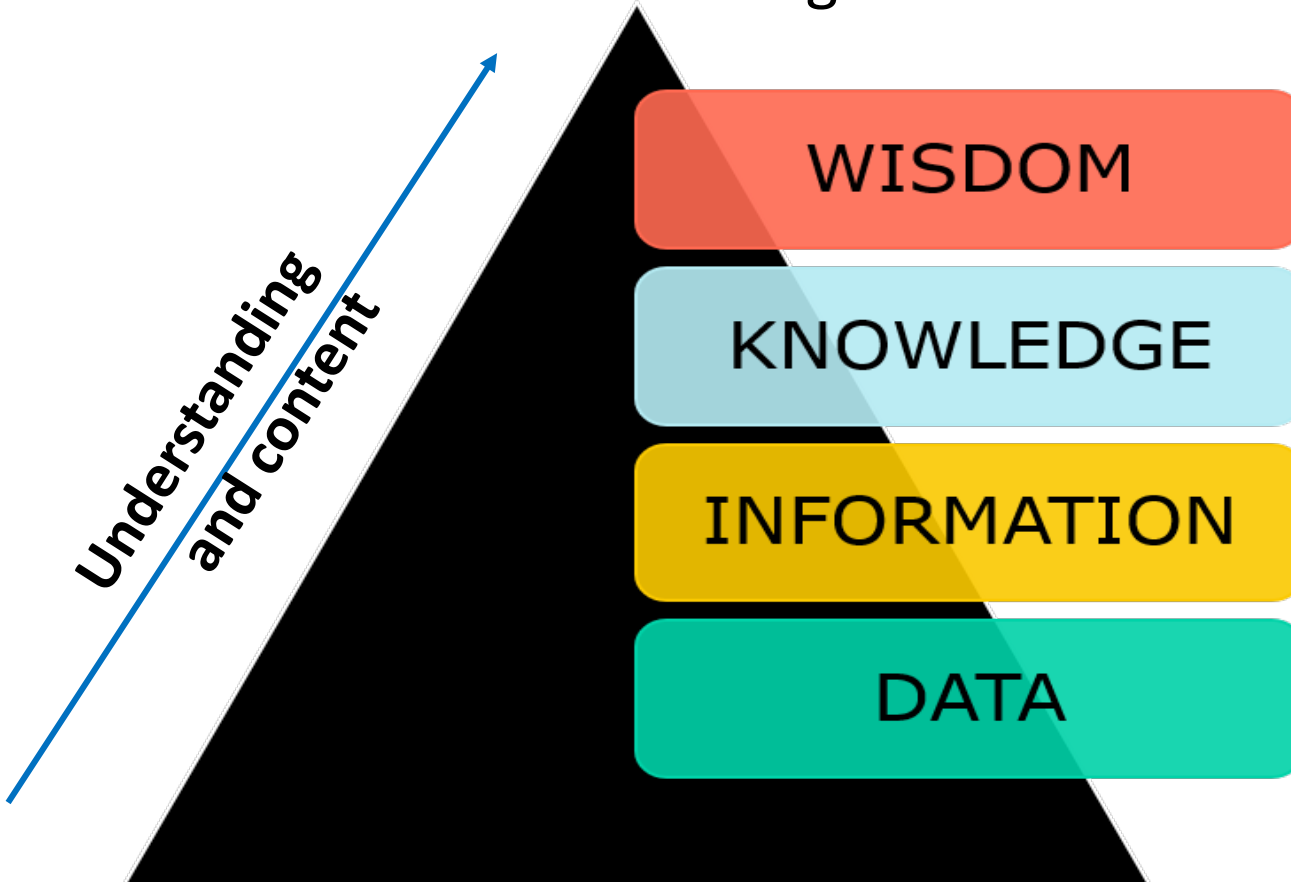
- Imitation of human learning



How AI Works

■ How does artificial intelligence learn?

- mimics human learning

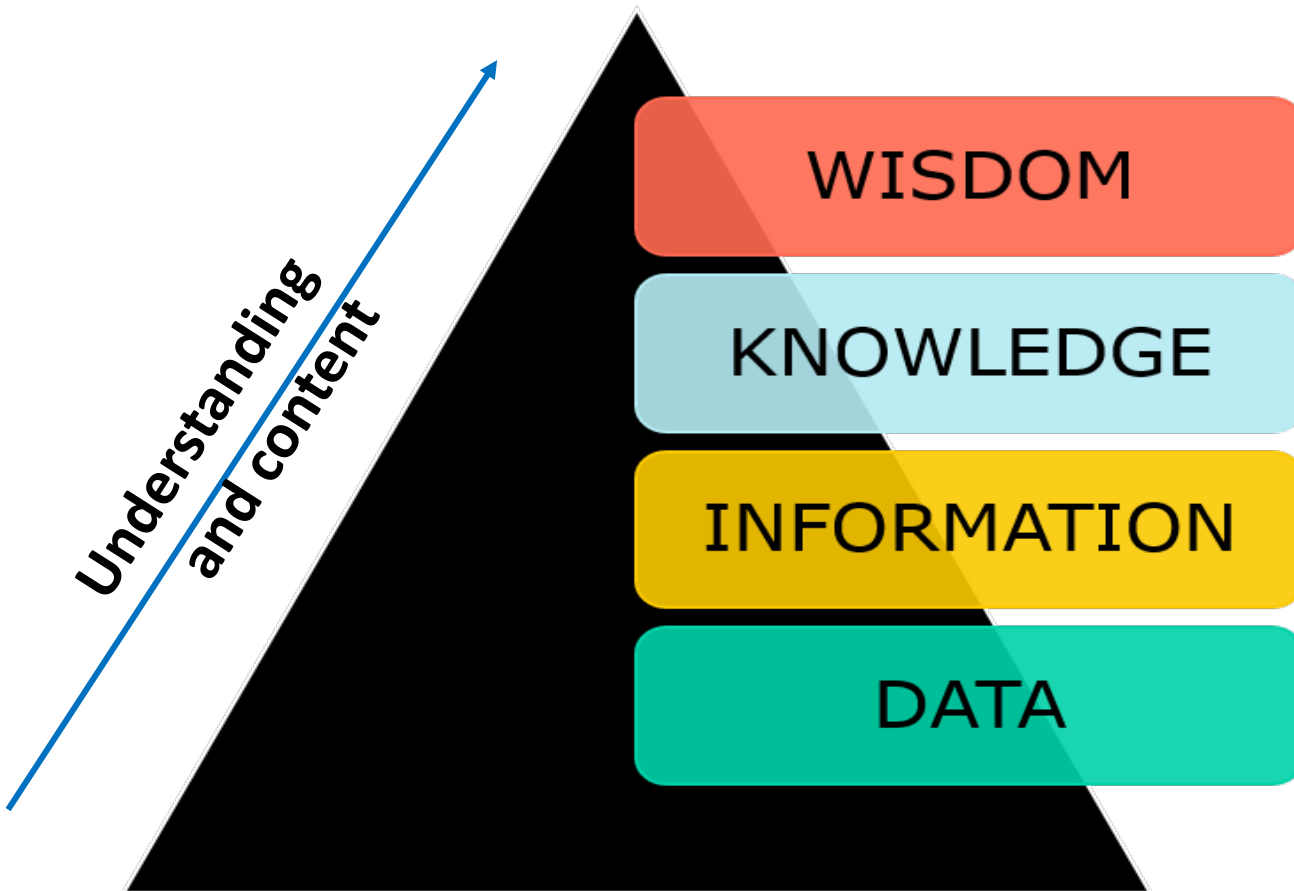


Data consists of raw, unstructured facts, similar to raw materials that need to be processed.

The data is like scattered pieces of "Lego" bricks, which in themselves do not constitute any meaning or benefit.

Example : Employees birth dates

How AI Works

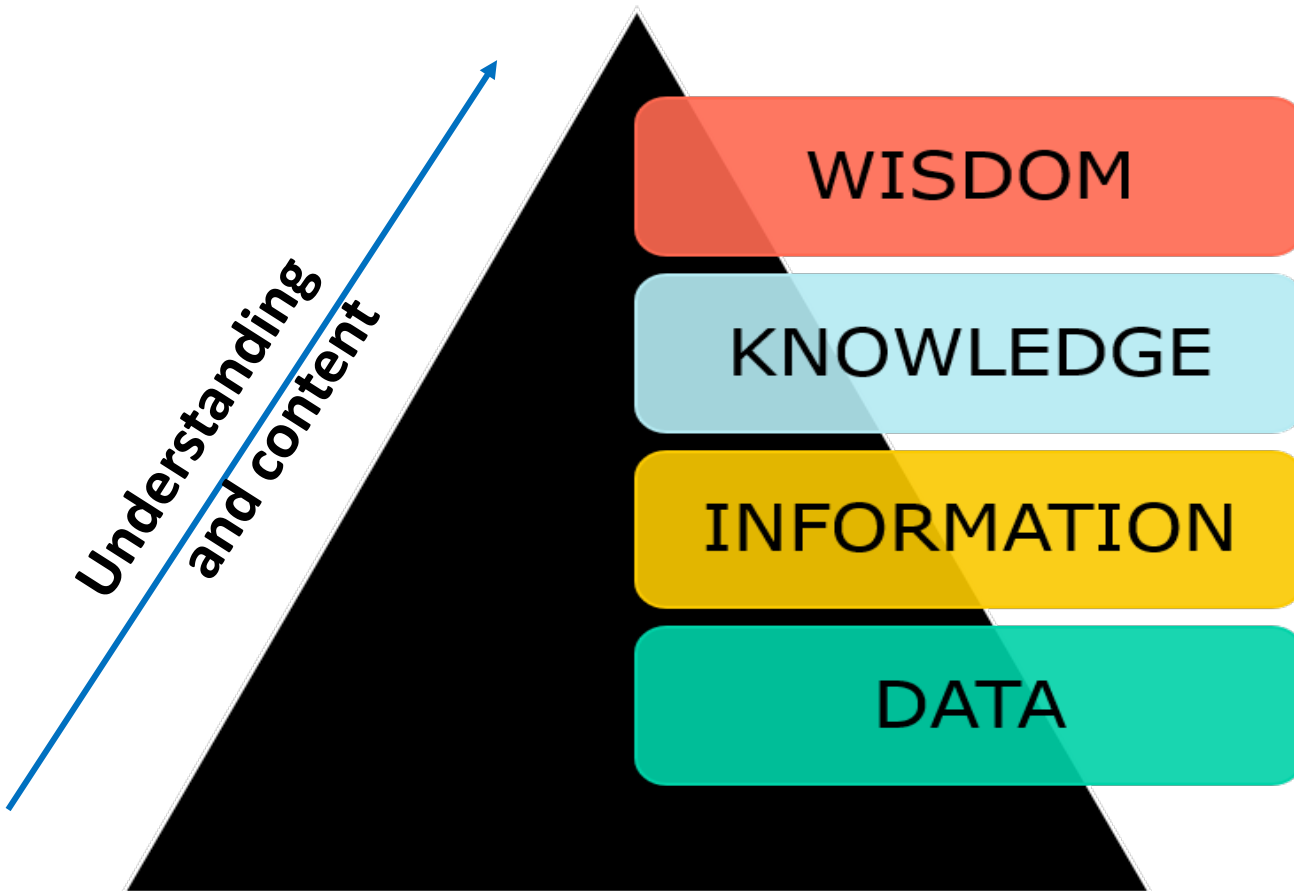


Information is organized and processed data, so that it carries a specific meaning and benefit.

It is the result of arranging and assembling Lego pieces to form meaningful shapes.

Example : Employees ages and their averages

How AI Works

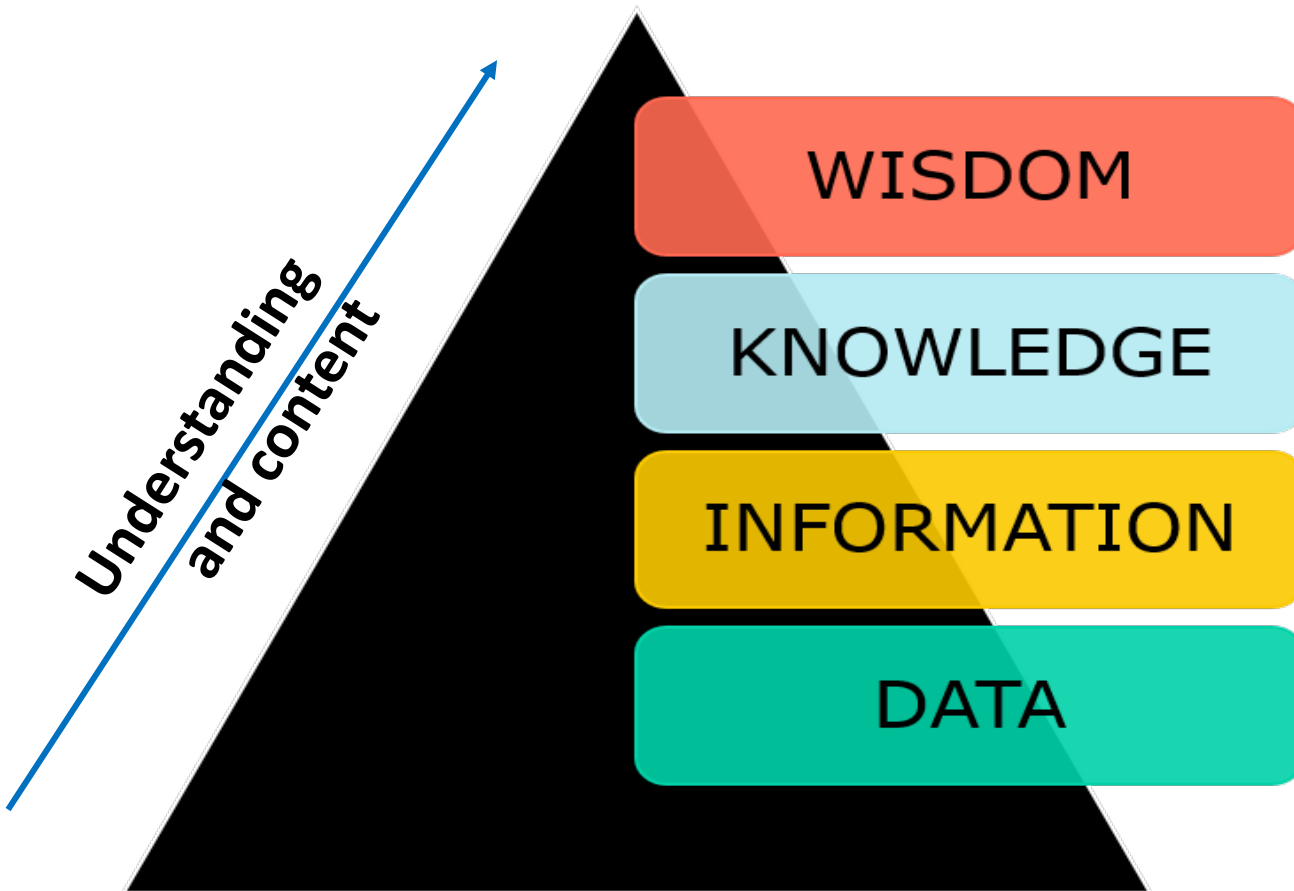


Knowledge is the product of applying, understanding, and using information practically and effectively.

This is the stage where we realize the complete picture formed by the "Lego" model.

Example : How to distribute work according to age average

How AI Works



Wisdom is the pinnacle of enlightened knowledge based on experience and sound judgment.

This is the stage where we become able to build new and innovative "LEGO" models.

Example : Making decisions about, transfer, and replacement policies.

How AI Works

■ Similarities between learning in artificial intelligence and human learning

- Artificial intelligence is originally inspired by the way humans learn.
- When we design a learning algorithm, we try to make it go through processes similar to those that the human brain goes through:

Observation → Comprehension → Repetition → Evaluation → Correction → Learning

- This is what makes artificial intelligence seem "intelligent".
- Because it follows the same general logic of human learning,
- But in the language of numbers and data, instead of experience and emotion.



How AI Works

■ Comparison between the stages of learning in humans and machines

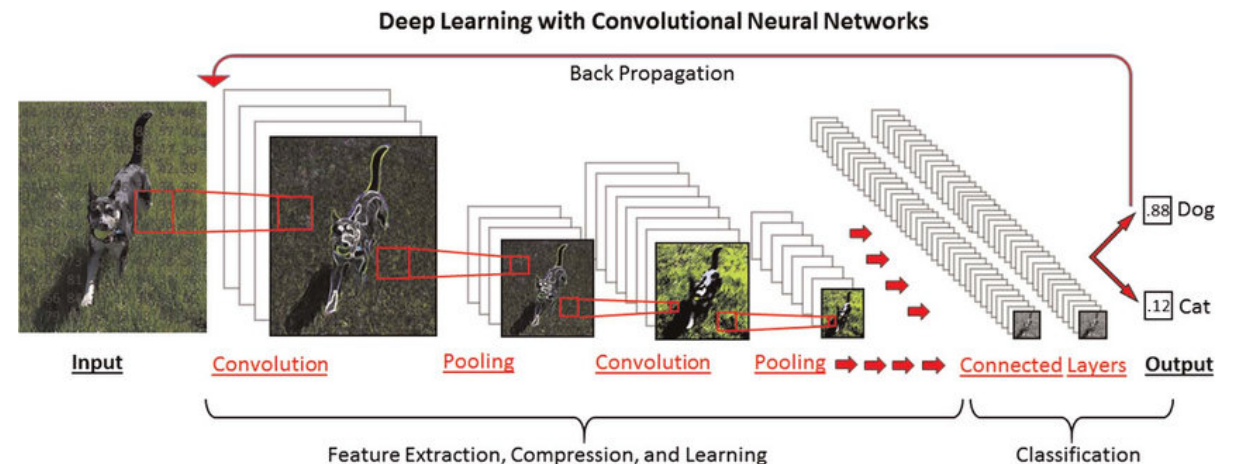
Stage	Learning in artificial intelligence	human learning
Inputs	The system collects data from images, texts, or sensors.	Humans see, hear, and read from the environment.
Memory	The system stores data in databases or neural network weights.	The brain stores experiences and situations in long-term and short-term memory.
Processing	The algorithm analyzes the data and looks for patterns and statistical relationships.	The brain connects new information with prior knowledge to infer meaning.
Error and correction Feedback	The model adjusts weights to reduce the error in prediction (backpropagation).	Humans learn from mistakes through experience and feedback.
Learning	The model improves as it receives new data.	Knowledge accumulates with practice.
Forgetting or bias	Artificial intelligence may be affected by data bias.	Humans forget or are influenced by their emotions.

How AI Works

■ Practical examples of the similarities between the learning stages in humans and machines

Example 1: The child learns to identify animals

- The child sees pictures of several cats and dogs, then gradually begins to distinguish between them.
- At first he makes a mistake (he calls a small dog a “cat”),
- But with correction, he develops his mental model.
- Just like an AI model does when training on categorized images.



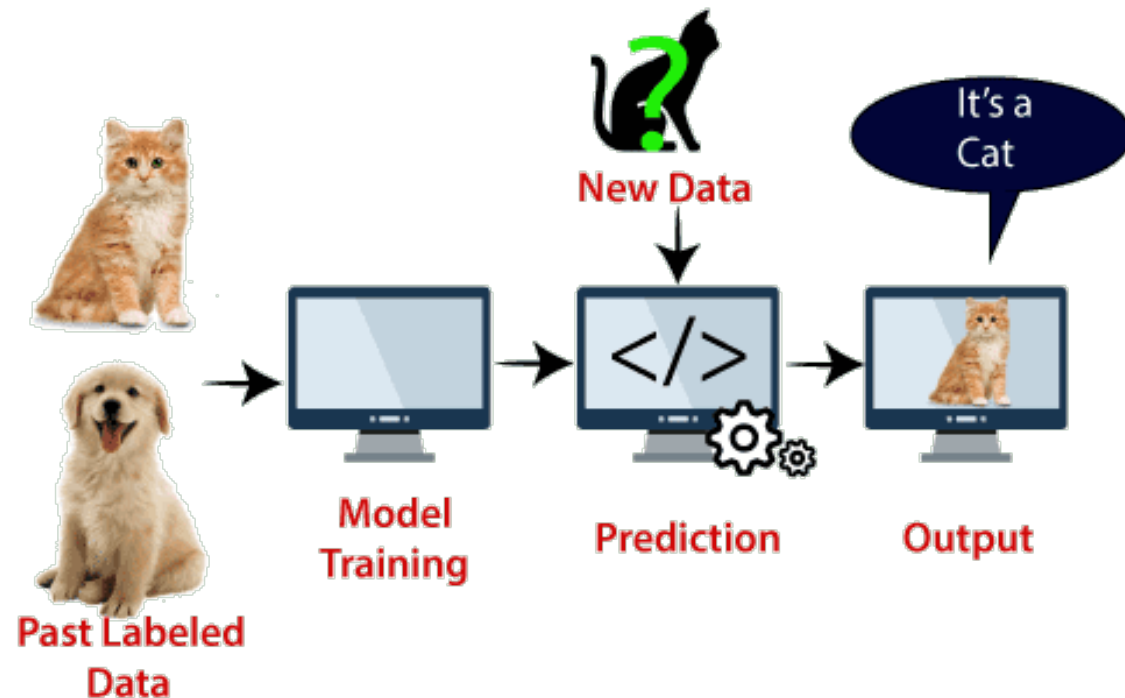
How AI Works

- Practical examples of the similarities between the learning stages in humans and machines

Example 1: The child learns to identify animals

- **Result** : Both the child and the model construct a mental pattern.

Pattern Recognition

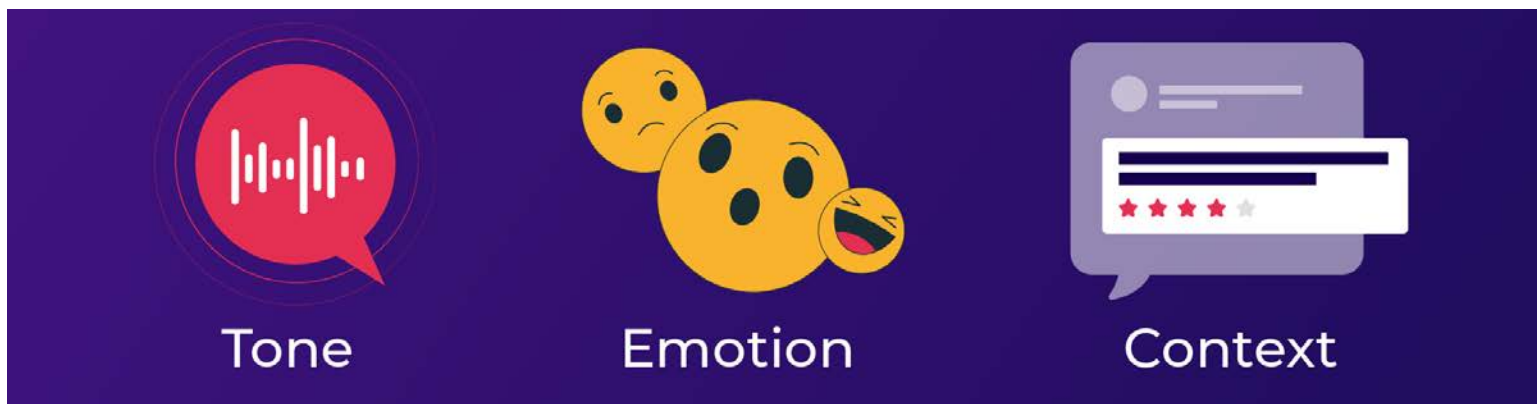


How AI Works

■ Practical examples of the similarities between the learning stages in humans and machines

Example 2: The media system learned to analyze emotions.

- Just as a journalist learns to understand the tone of the audience (angry, sarcastic, supportive) from words, Artificial intelligence learns by analyzing thousands of tweets and reviews.
- Over time, he becomes capable of "reading the general mood" with human-like accuracy.

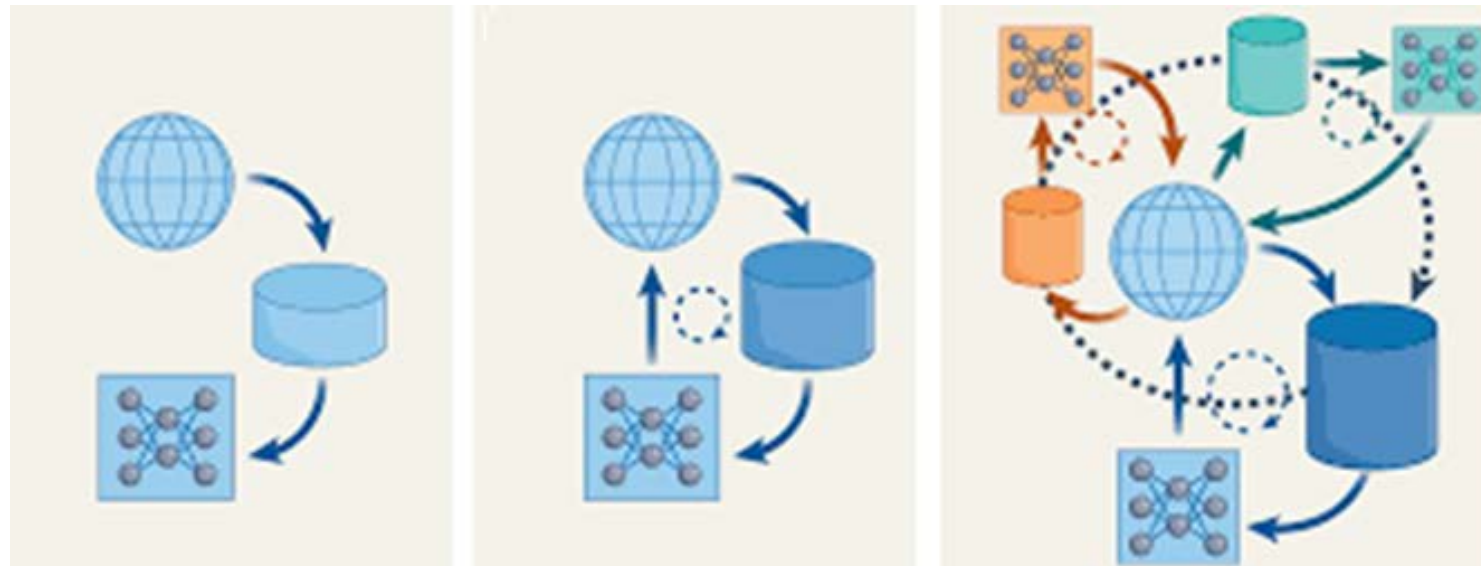


How AI Works

■ Practical examples of similarity

Example 3: Practice leads to improvement

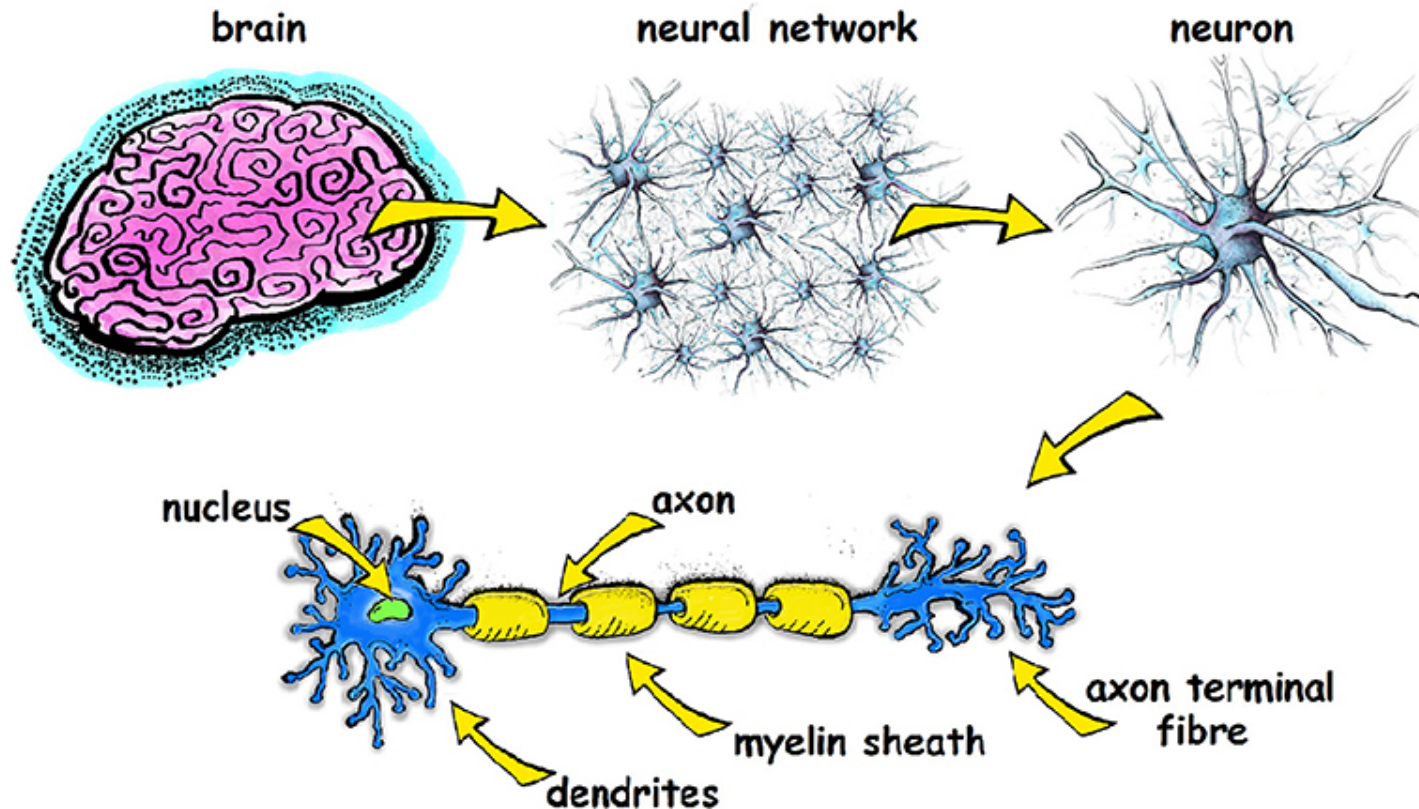
- Humans improve their performance through training and continuous observation
- Artificial intelligence improves its performance through "retraining" on more recent data.



How AI Works

■ Scientific background: Similarity in structure as well

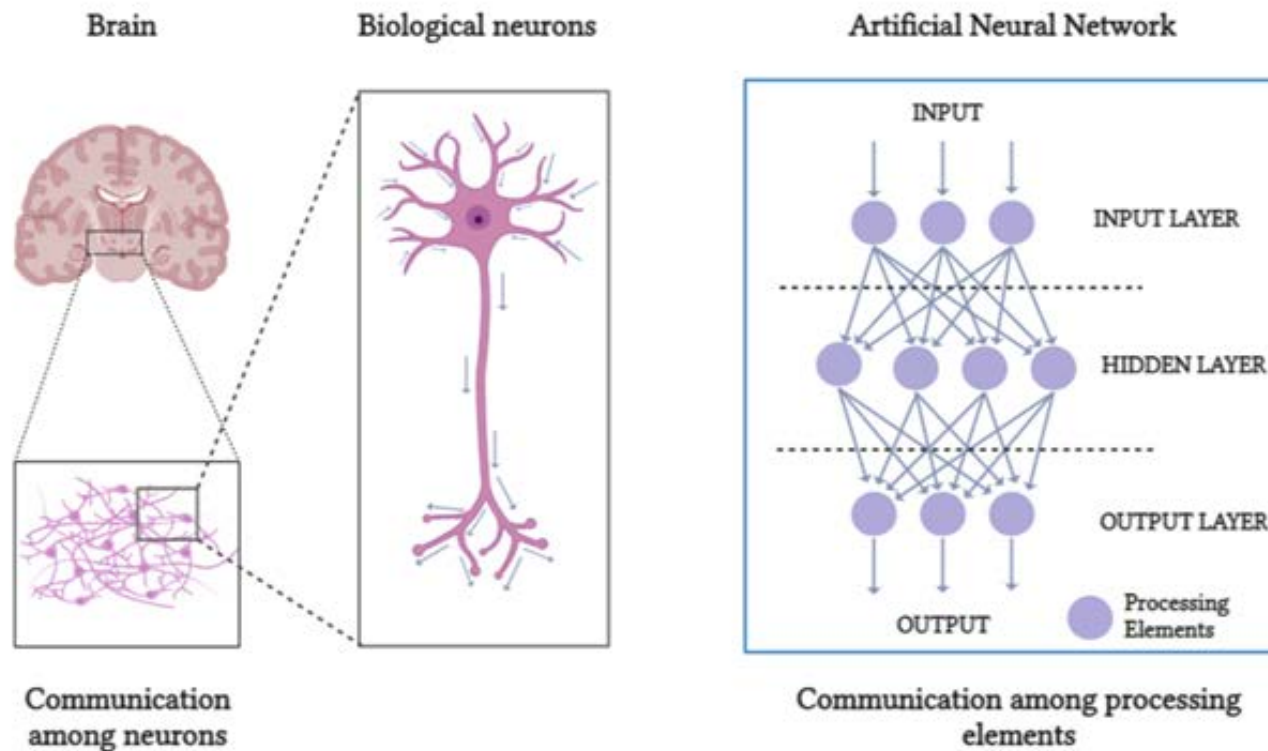
- The human brain contains a network of nerve cells. Neurons exchange electrical signals.



How AI Works

■ Scientific background: Similarity in structure as well

- Artificial intelligence uses artificial neural networks (ANN), which are mathematical representations that mimic the workings of those cells.

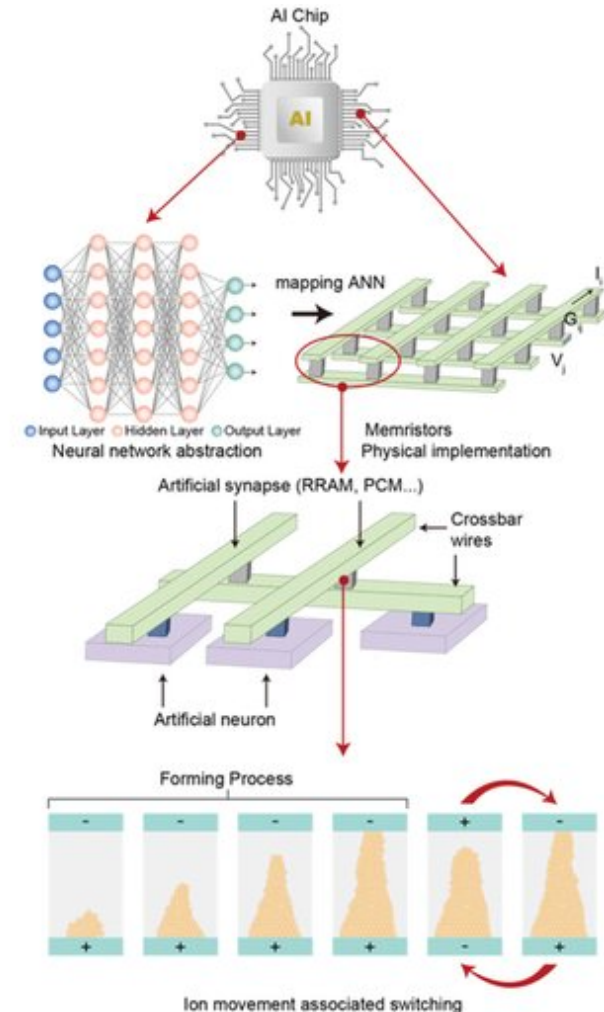
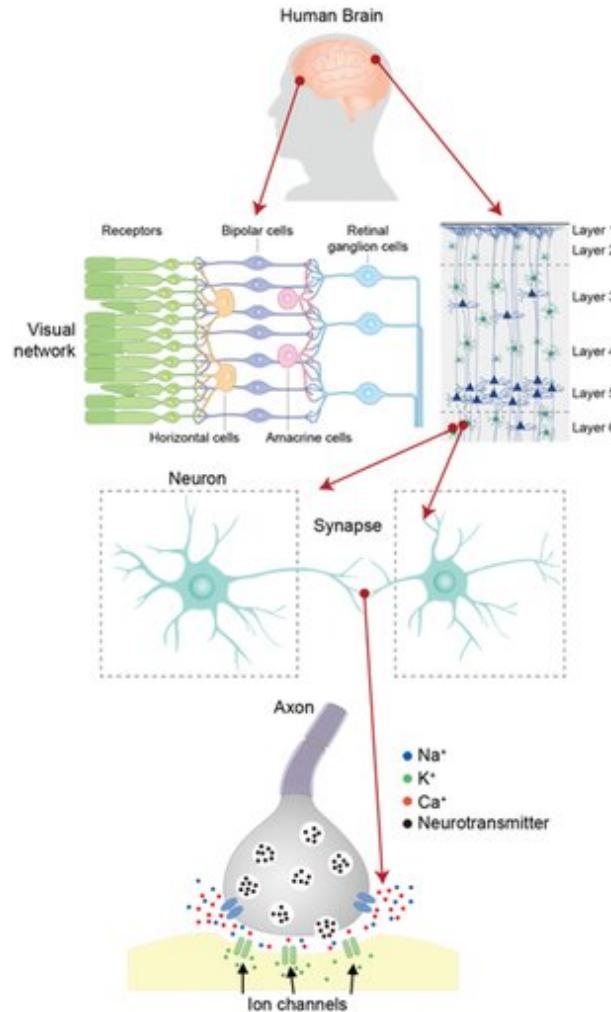


How AI Works

■ Scientific background: Similarity in structure as well

In each artificial neuron:

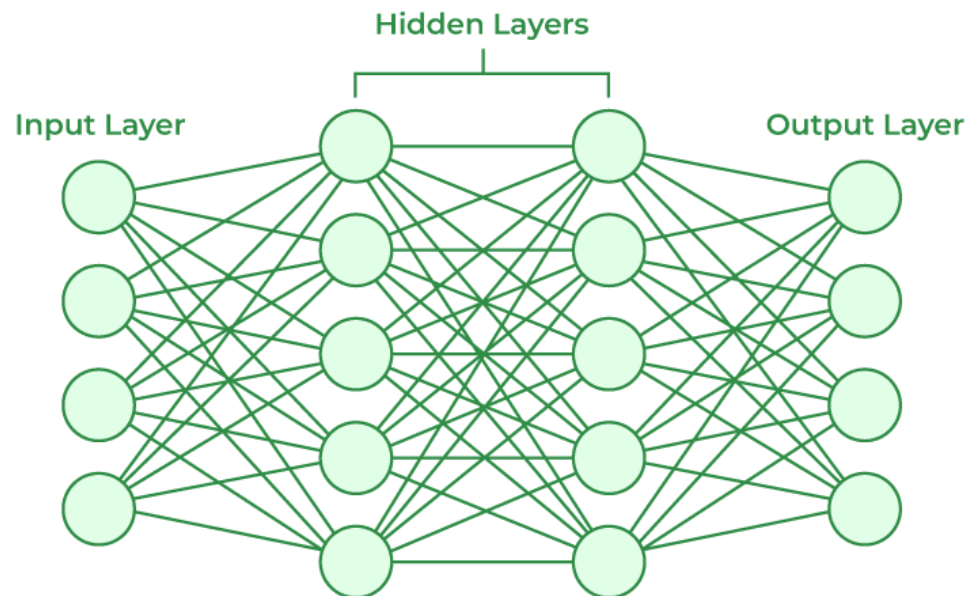
- **Inputs:** Signals from previous layers.
- **Processing operation:** Multiplying values by weights
- **Activation:** A signal that passes if it exceeds a certain limit.
- Just like what happens in the brain when making a decision or realizing new information.



How AI Works

■ A simplified explanation of artificial neural networks

- It was designed to mimic the way nerve cells in the brain work.
- **Inputs:** Raw data (images, texts, numbers)
- **Hidden Layers :** Process data and extract patterns
- **Outputs :** The final result (prediction, decision, classification)

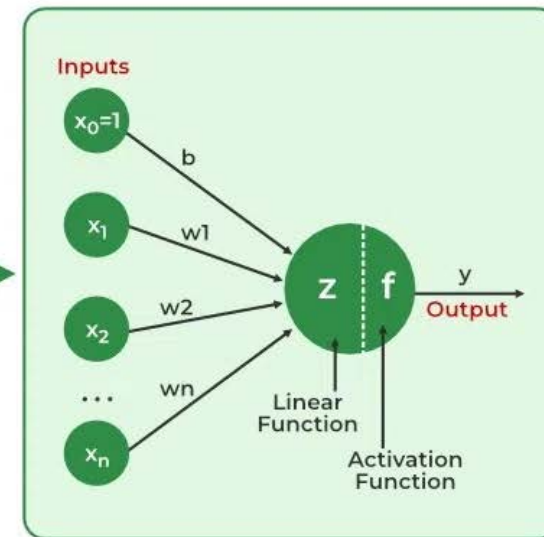
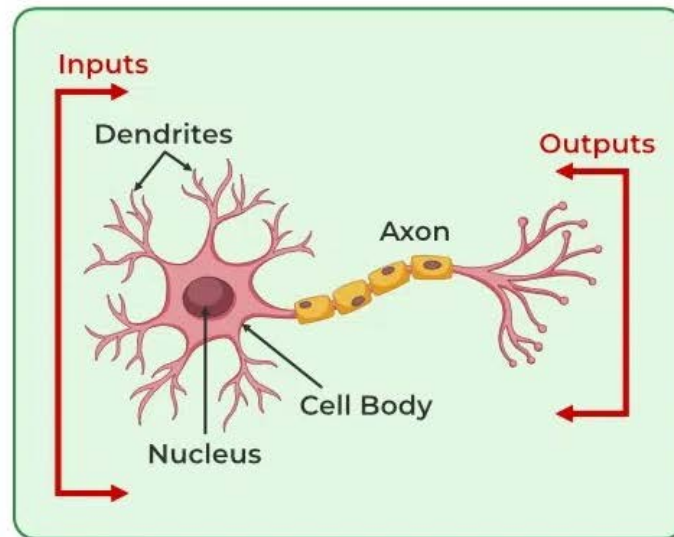


How AI Works

■ Artificial neuron (Perceptron)

- simplest type of neural network
- basis for building more complex models .
- It is a machine learning algorithm used to learn from classified training data .
- It is also known as a single-layer neural network and is typically used for classification tasks that distinguish between two categories of data.

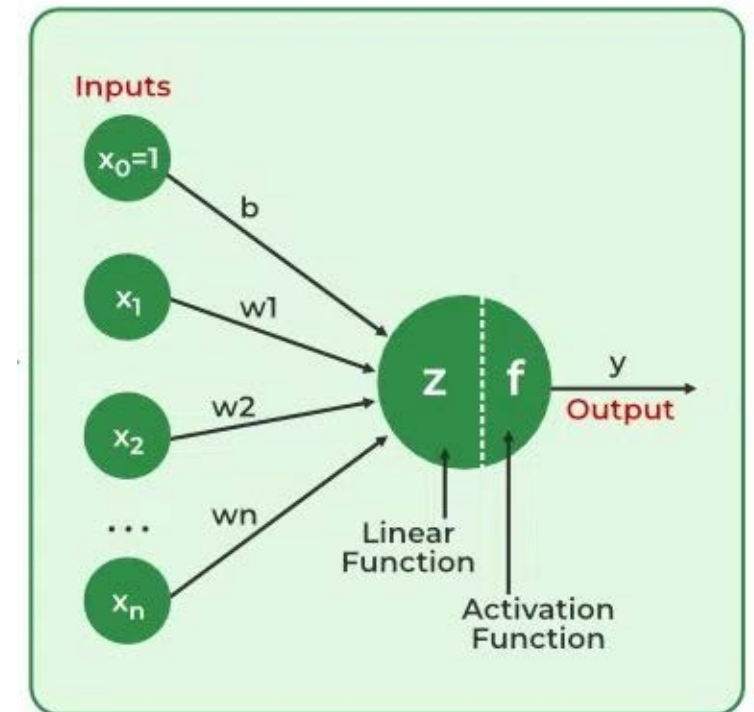
To visualize the perceptron, it can be thought of as a single artificial neuron.



How AI Works

■ How an artificial neuron (perceptron) works

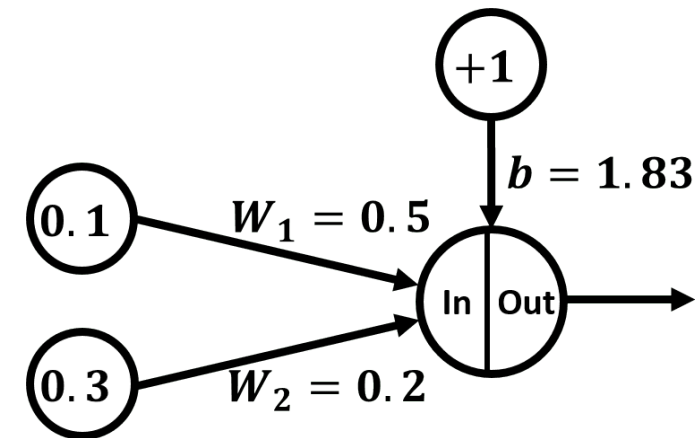
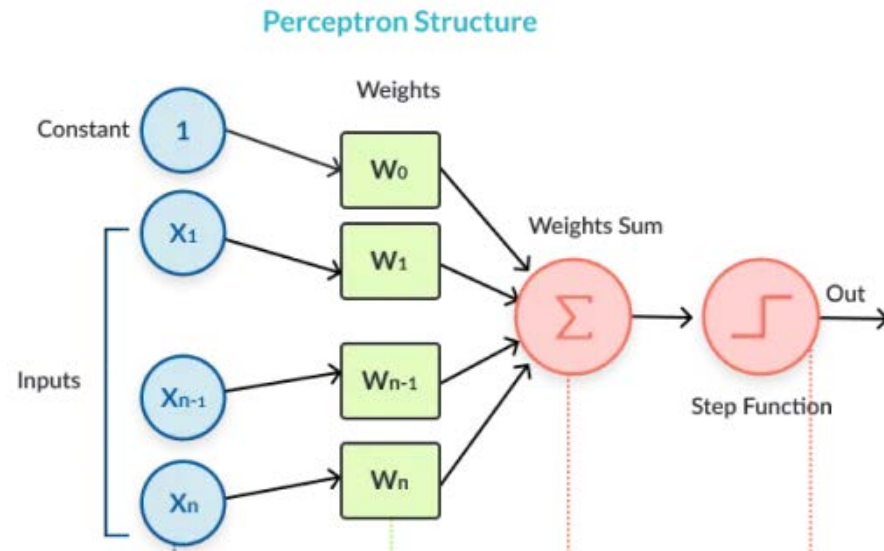
- The perceptron receives several inputs, combines them with weights, decides which category they belong to, and acts as a simple decision-maker.
- It consists of four main parameters:
 - Input values (also called nodes),
 - weights
 - biases
 - Net Sum
 - Activation



How AI Works

■ Artificial neuron :

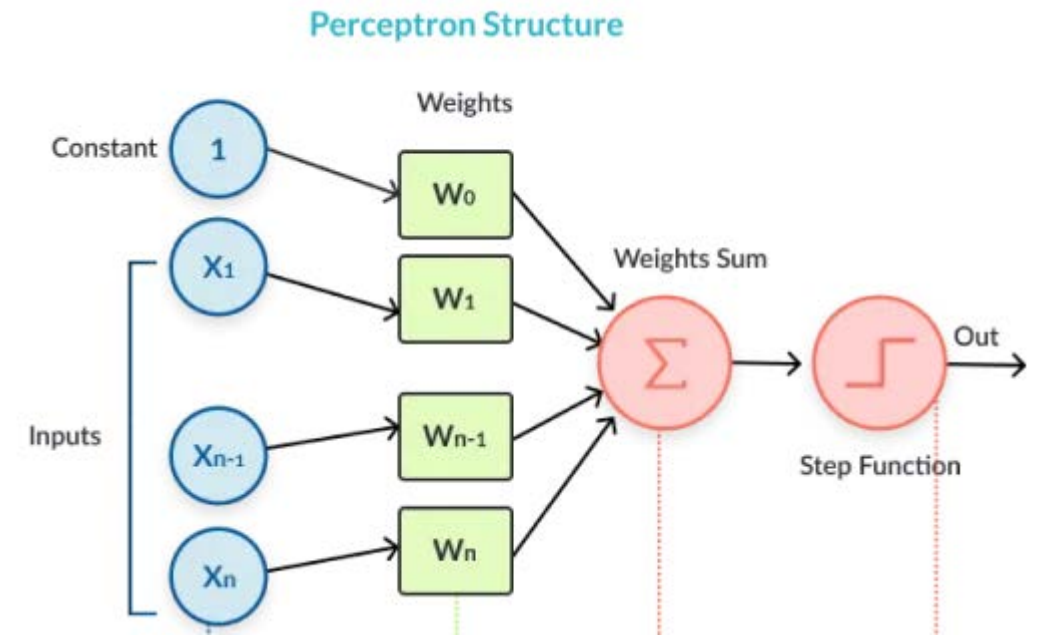
- **Inputs and weights:** There are pieces of information (inputs), and each has a weight that indicates its importance.
- **Bias** helps the neuron to be more flexible in decision-making.
- **Calculating the weighted sum:** The neuron multiplies each input by its weight, then adds everything together, including bias, to arrive at a weighted sum.



How AI Works

■ Artificial neuron :

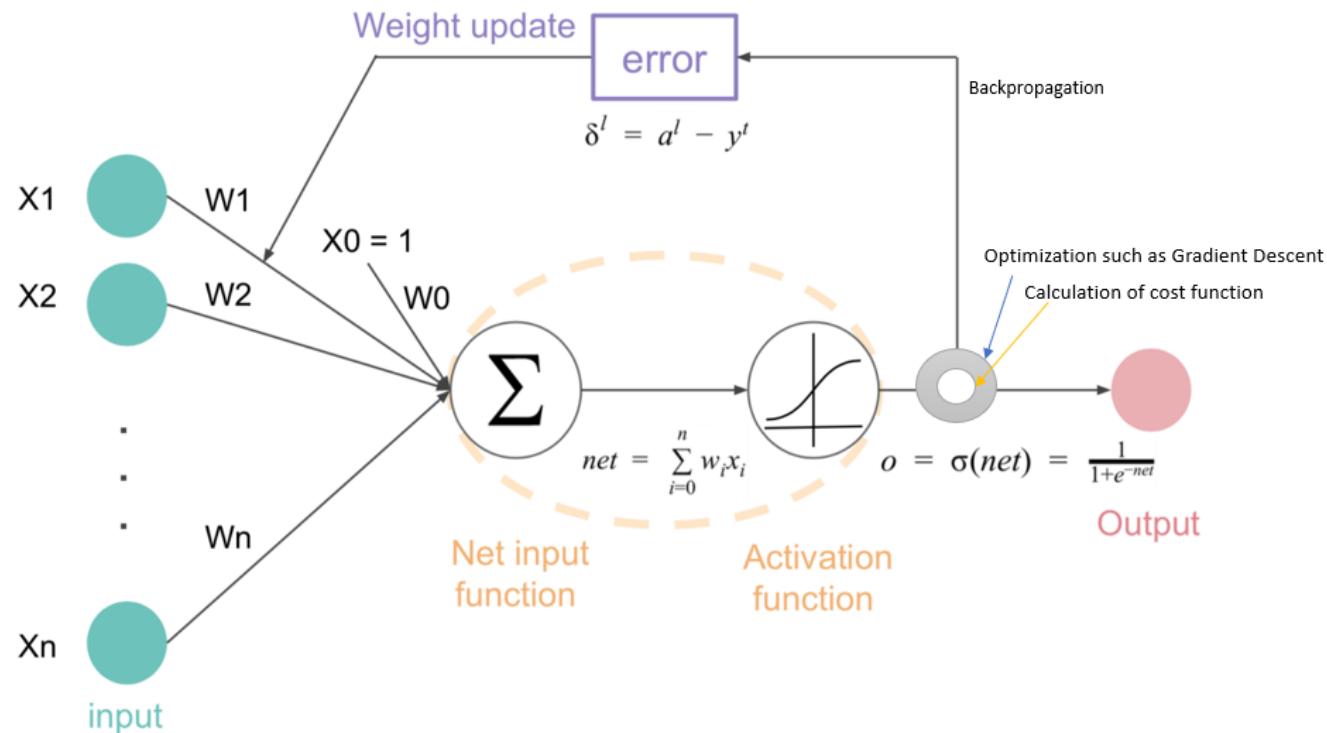
- **Making a decision:** Next, the perceptron uses an activation function to determine the final output.
- The weighted sum is taken and a decision is made as to whether the perceptron should be activated or not.
- In its simplest form, the activation function is a step function that yields a result of one if the weighted sum is above a certain threshold, and zero if it is not. Therefore, the weighted input is converted into a yes or no decision.



How AI Works

■ Artificial neuron :

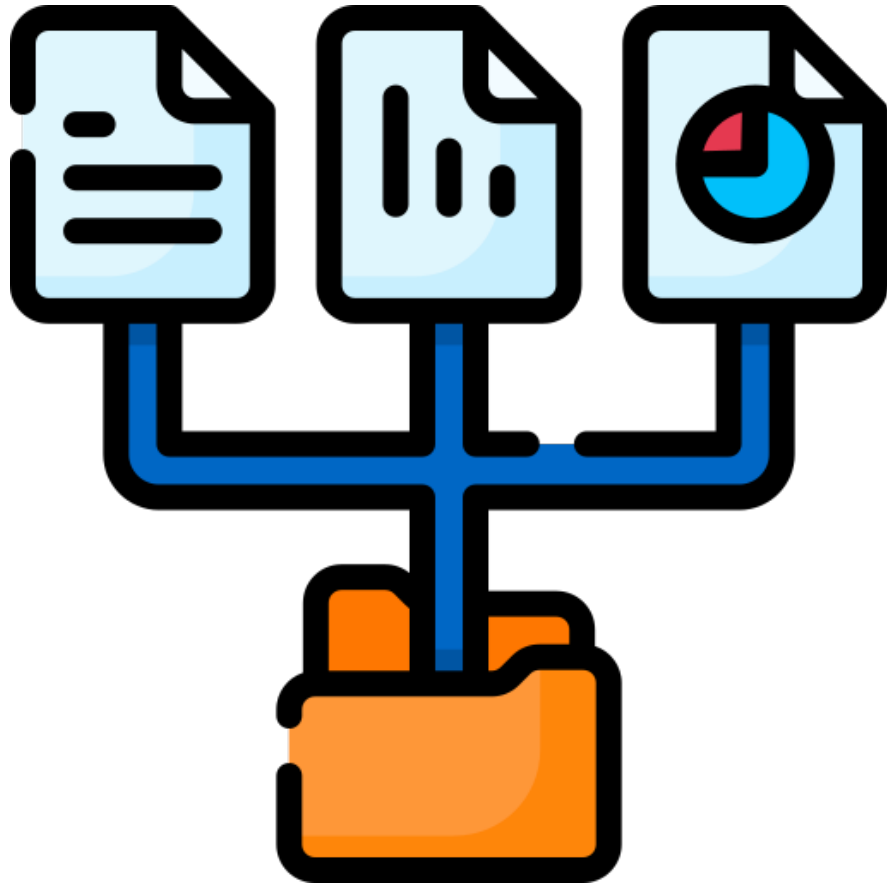
- **Learning and improvement:** If the perceptron makes a mistake, it adjusts the weights to try and improve next time. This process helps the model learn from its errors.



How does artificial intelligence learn?



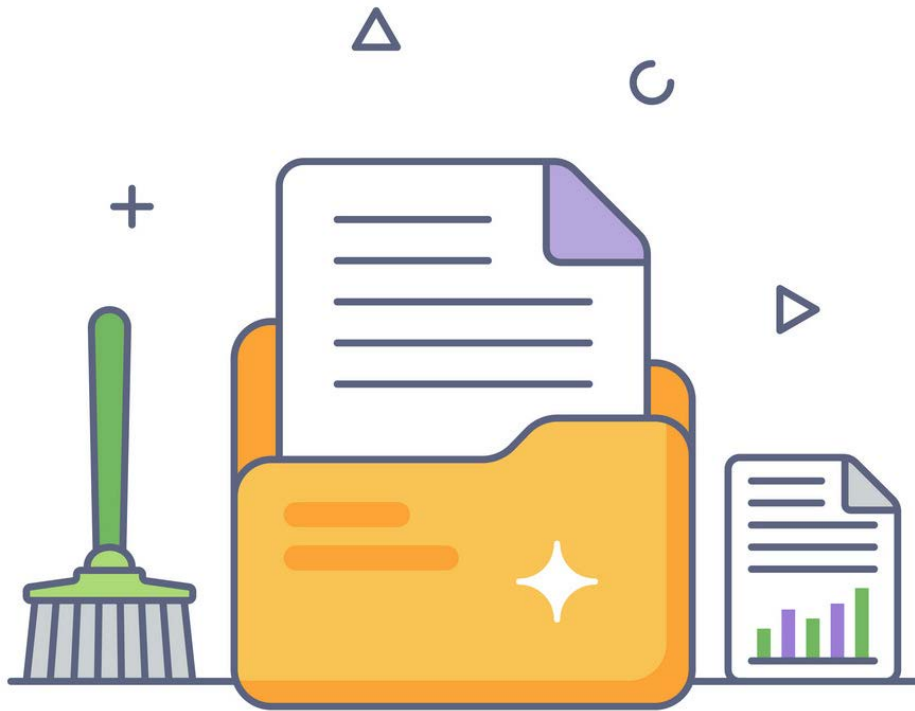
How does artificial intelligence learn?



■ Step 1: Data Collection

- The system needs large amounts of information.
- It may be collected from: the internet, databases, cameras, social networks, or sensors.

How does artificial intelligence learn?



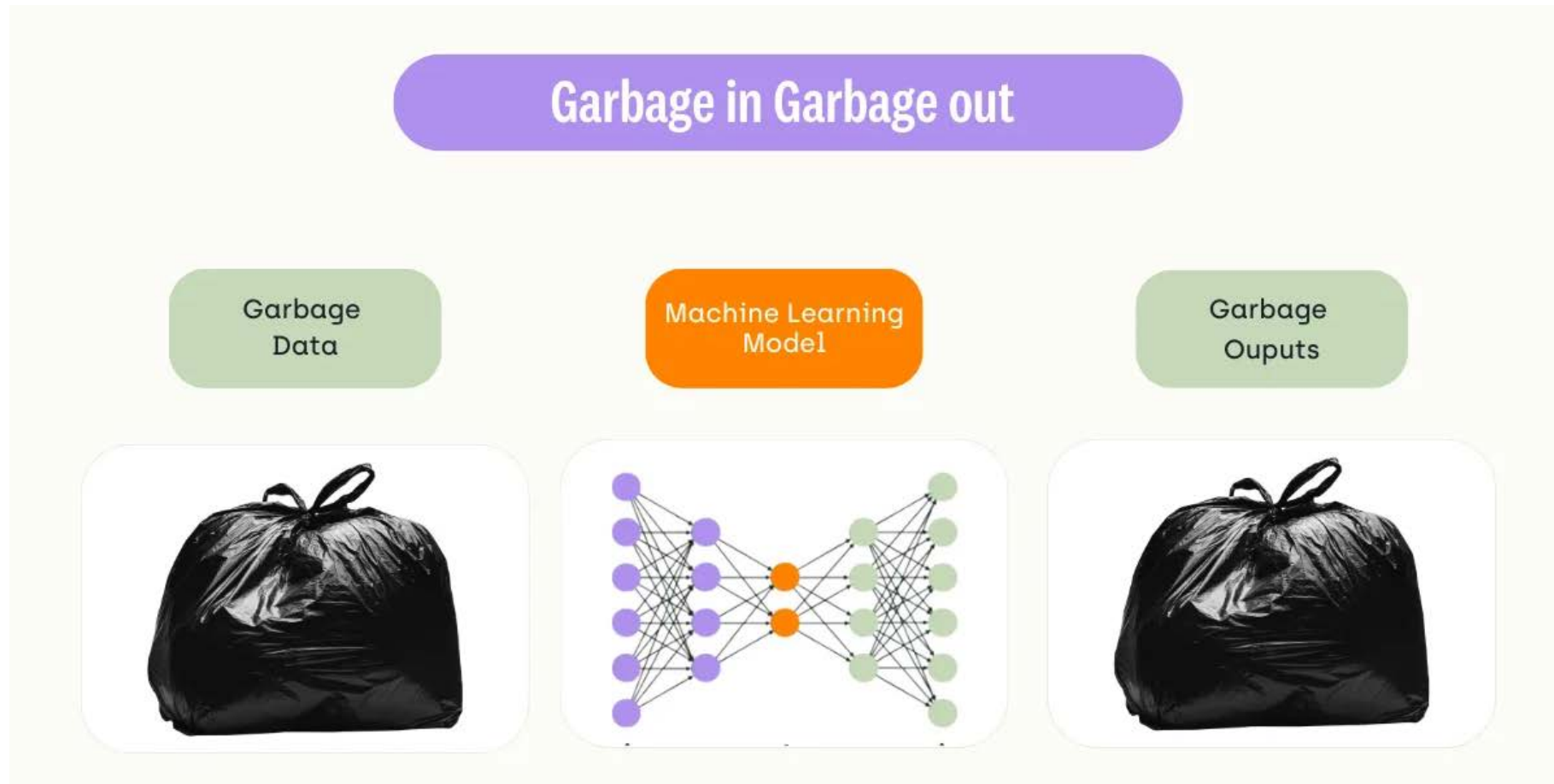
Data Cleaning

■ Step 2: Data preparation

- Cleaning the data from errors or unimportant information.
- Data preparation or data cleaning is the process of sorting and filtering raw data to remove unnecessary, inaccurate, and repeated information.

How does artificial intelligence learn?

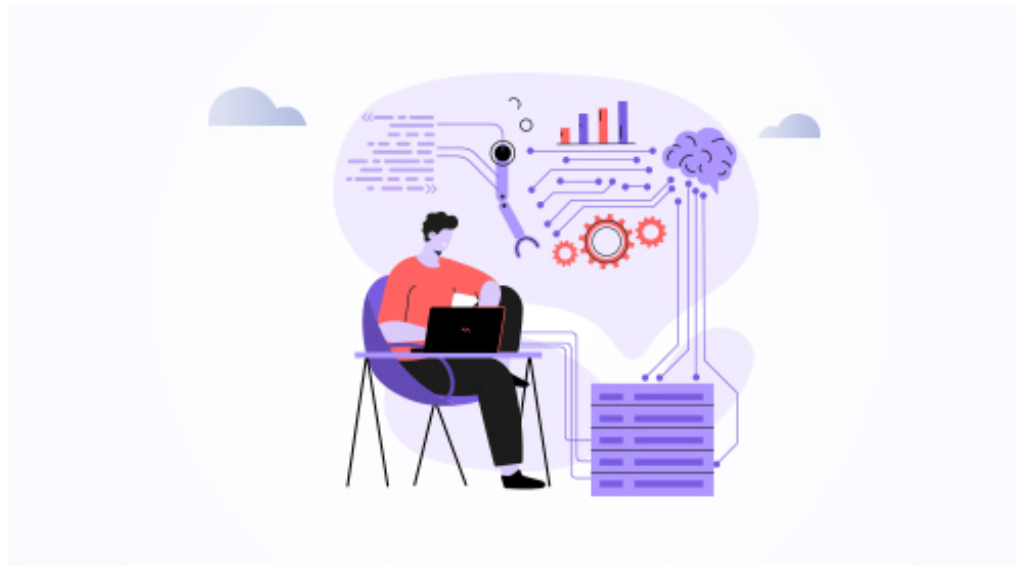
■ Step 2: Data preparation



How does artificial intelligence learn?

■ Step 3: Data Preprocessing

- Converting them into a form that algorithms can understand (such as numbers and vectors).
- **Example**
If the data includes text posts from Twitter, the system converts the words to numbers using techniques such as Word Embedding .



How does artificial intelligence learn?

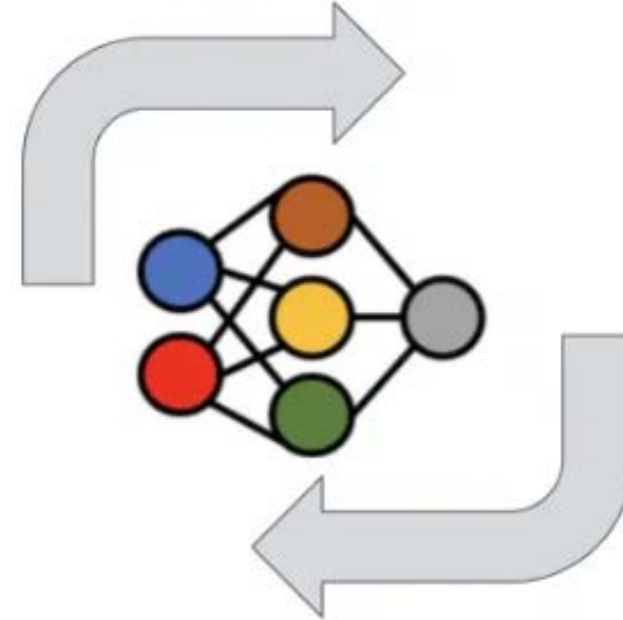
■ Step 4: Training

- Data is entered into a learning algorithm so that the system can begin to detect patterns.
- The "weights" are adjusted internally to achieve the best possible accuracy.

TRAINING DATA



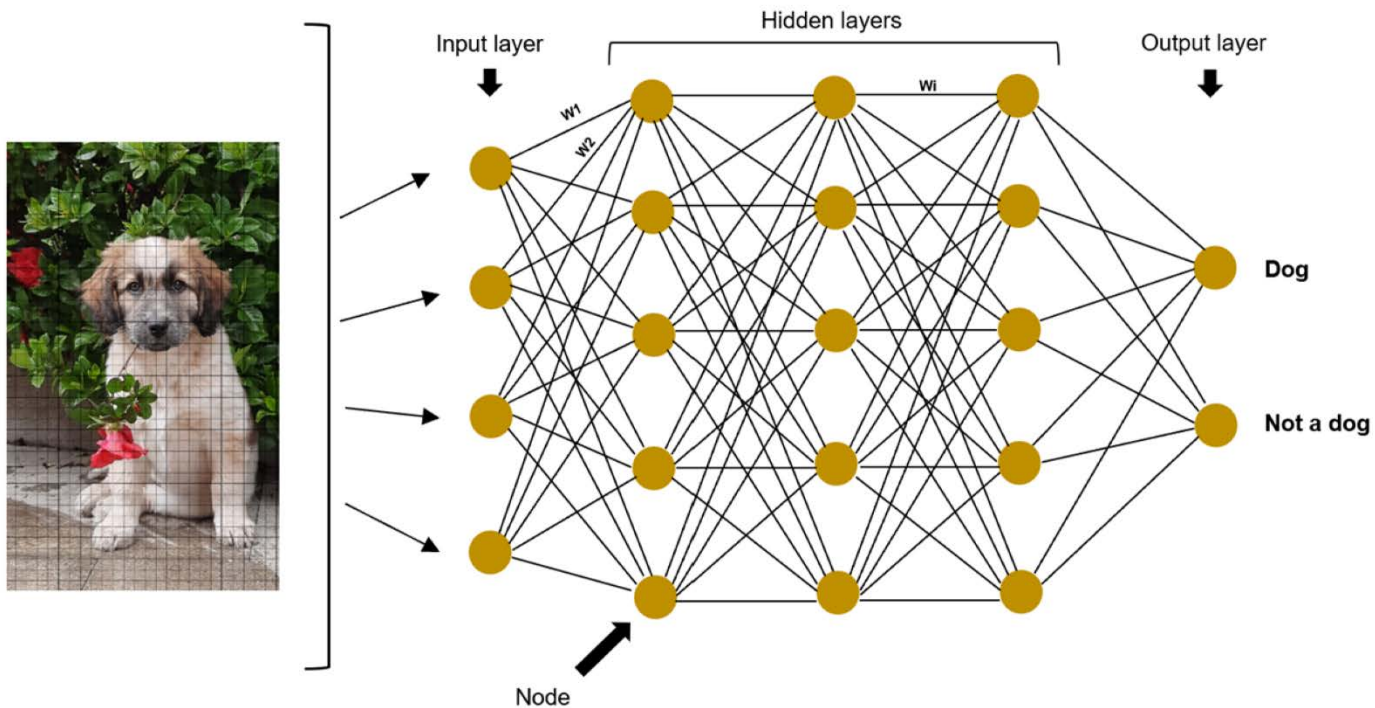
MODEL



How does artificial intelligence learn?

■ Step 4: Training

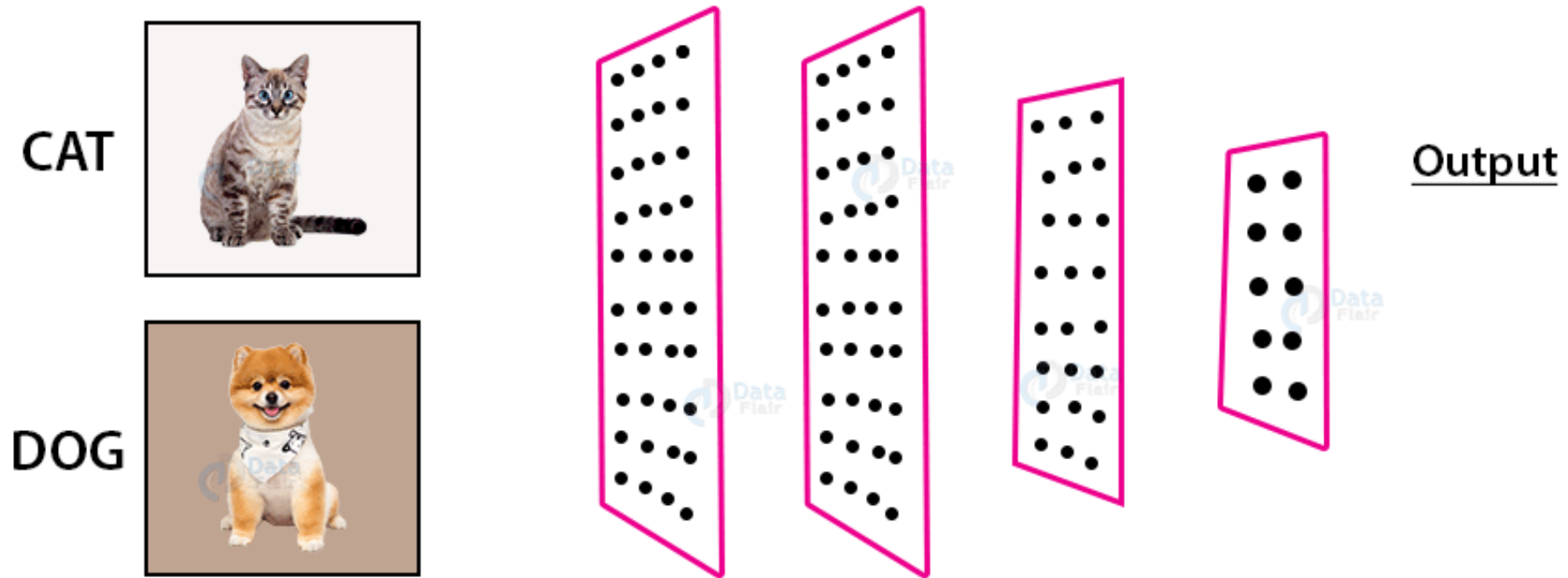
- Data is entered into a learning algorithm so that the system can begin to detect patterns.
- The "weights" are adjusted internally to achieve the best possible accuracy.



How does artificial intelligence learn?

■ Step 5 : Prediction or Decision Making

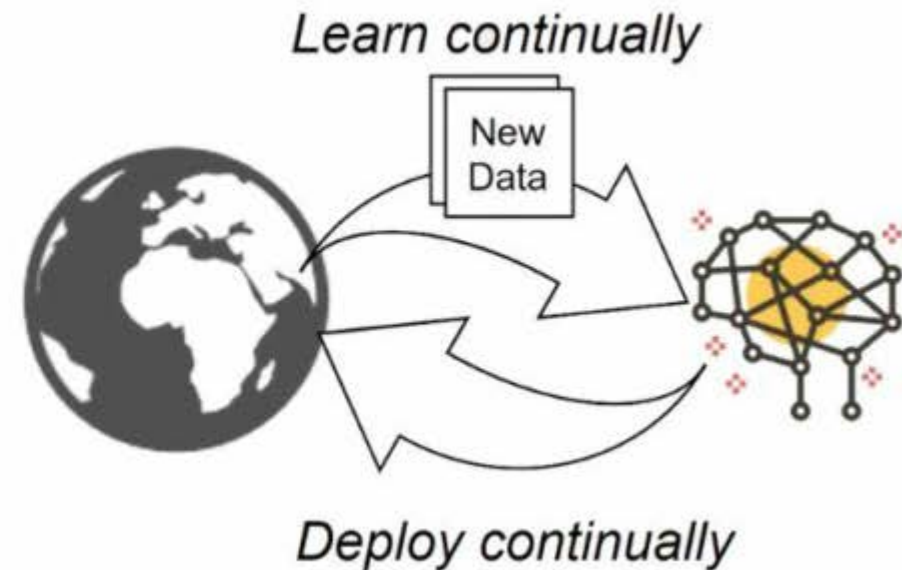
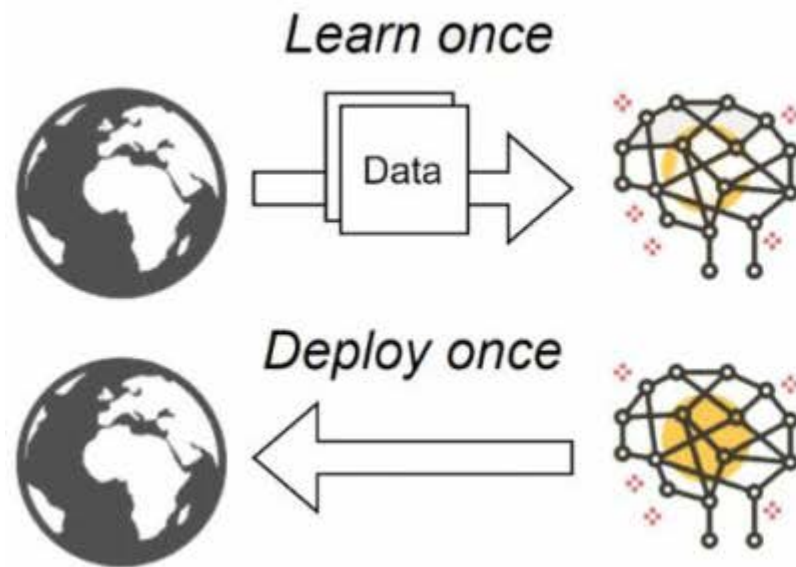
- After training, the system can now receive new data and make decisions or predictions.



How does artificial intelligence learn?

■ Step 6 : Continuous Learning

- Artificial intelligence never stops learning.
- As it receives new data, it improves itself and becomes more accurate over time.

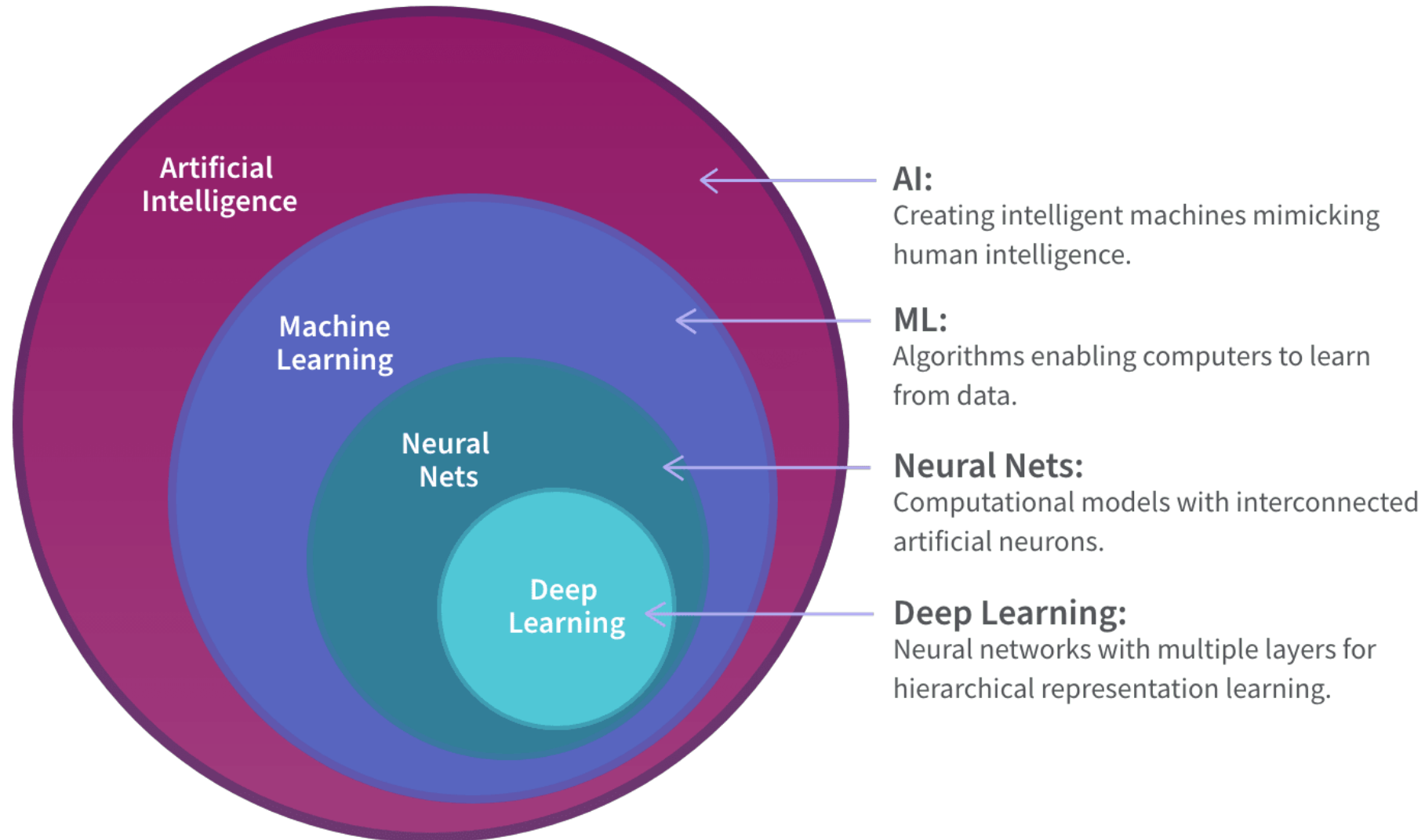


Types of learning in artificial intelligence

Type of learning	Definition	Example
Supervised Learning	The system learns from pre-classified data (with correct examples).	A model that learns to distinguish between regular mail and spam.
Unsupervised Learning	The system itself searches for patterns and similarities in unclassified data.	Analyzing audience segments without pre-defined labels.
Deep Learning	It relies on “artificial neural networks” that mimic the human brain.	Highly accurate facial or voice recognition.
Reinforcement Learning	The system learns from “right and wrong” through reward and punishment.	Robots and video games, or recommending ads based on interaction.



Types of learning in artificial intelligence

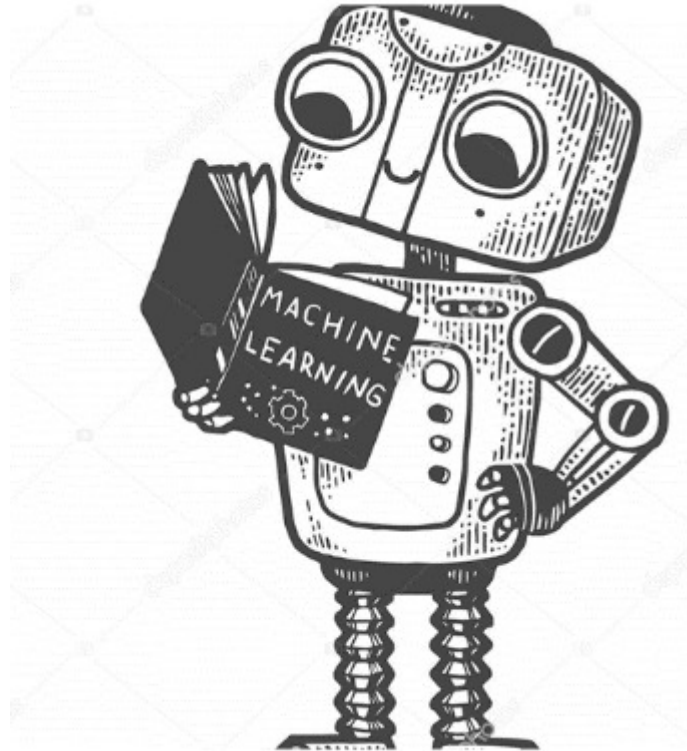


Machine learning and deep learning

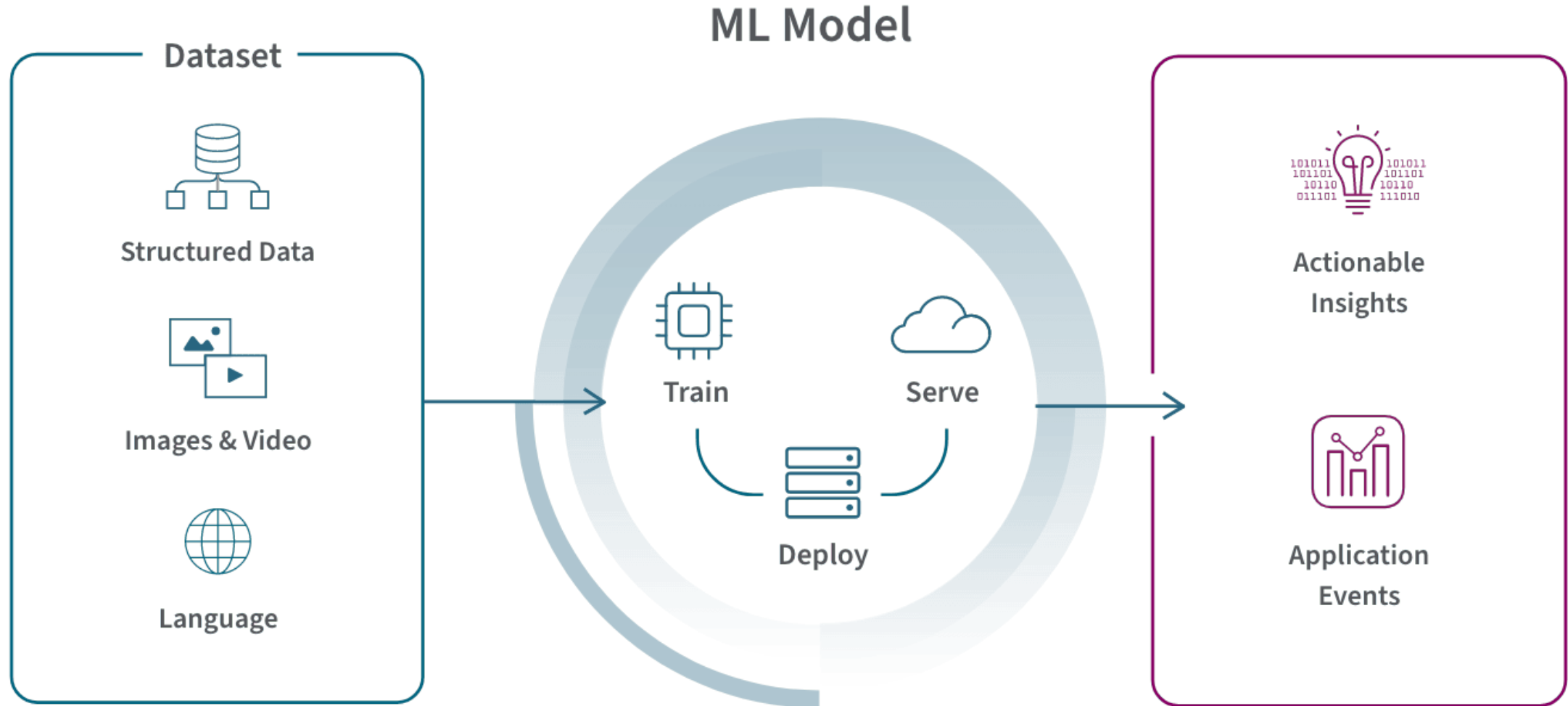
concepts and applications

■ Machine learning

- **Definition** : Making the machine learn from data instead of explicit instruction.



Types of learning in artificial intelligence





Types of learning in artificial intelligence

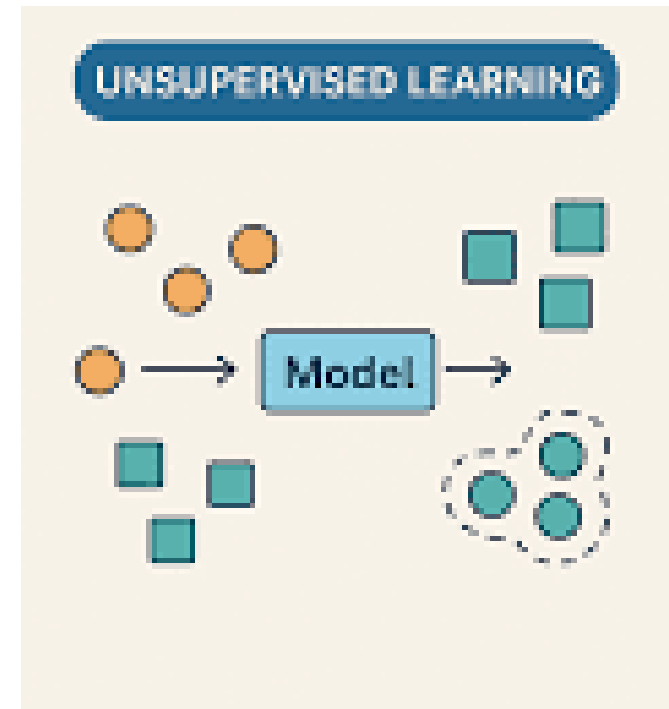
■ Types of learning:

1. Supervised learning
2. Unsupervised learning
3. reinforcement learning
4. Deep learning
5. Semi-Supervised learning
6. Self-Supervised Learning
7. Transfer learning

ML and DL – concepts and applications

■ Unsupervised Learning

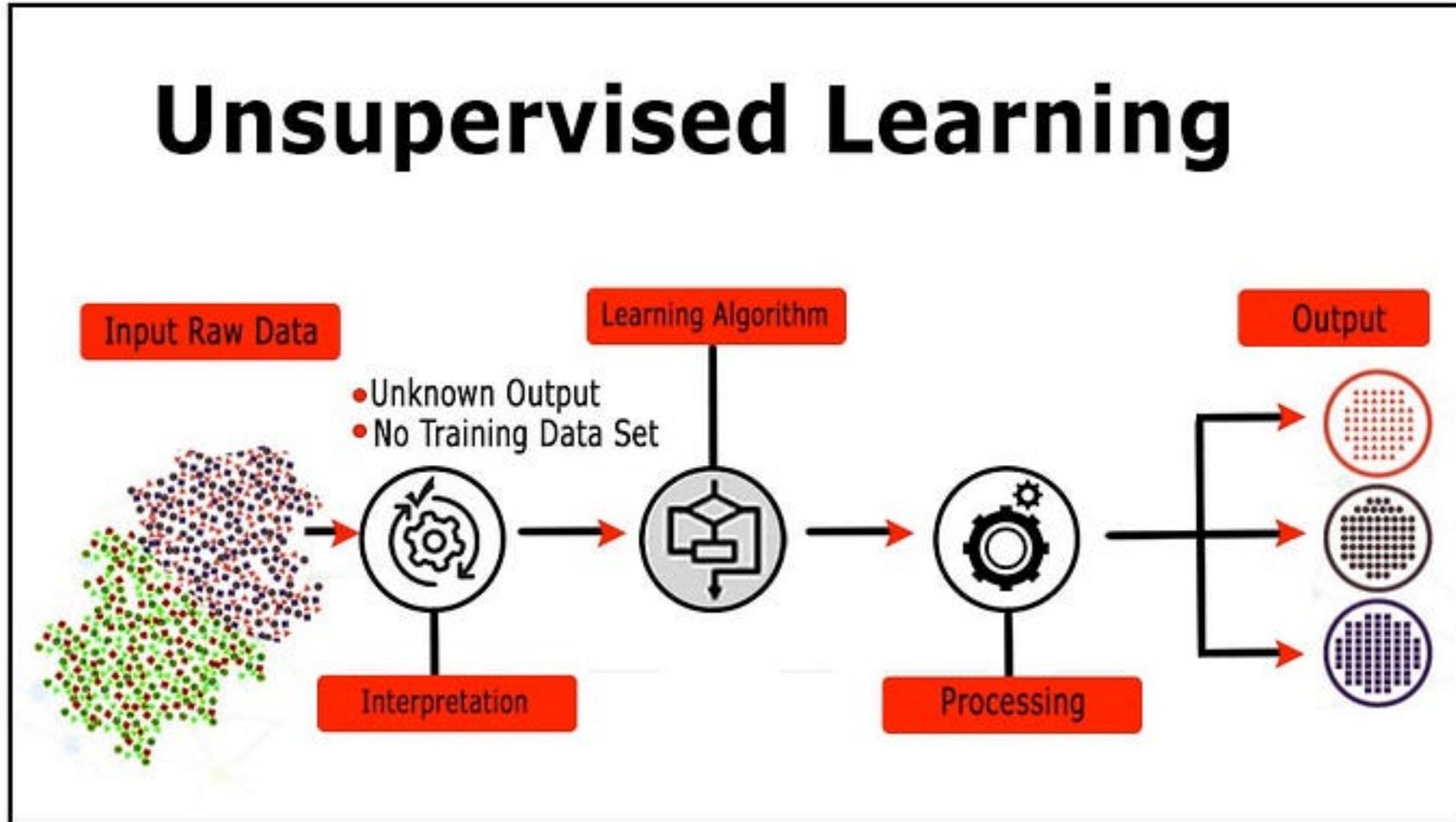
- This type of model is used to group data based on patterns and similarities between inputs, without any guidance from the output data.
- The goal is to use the model to discover natural patterns in the data, such as grouping the data or discovering its internal structure.



No labels, just looking for patterns in data

E.g. clustering, in data with multiple clusters, what are they, how big, etc.

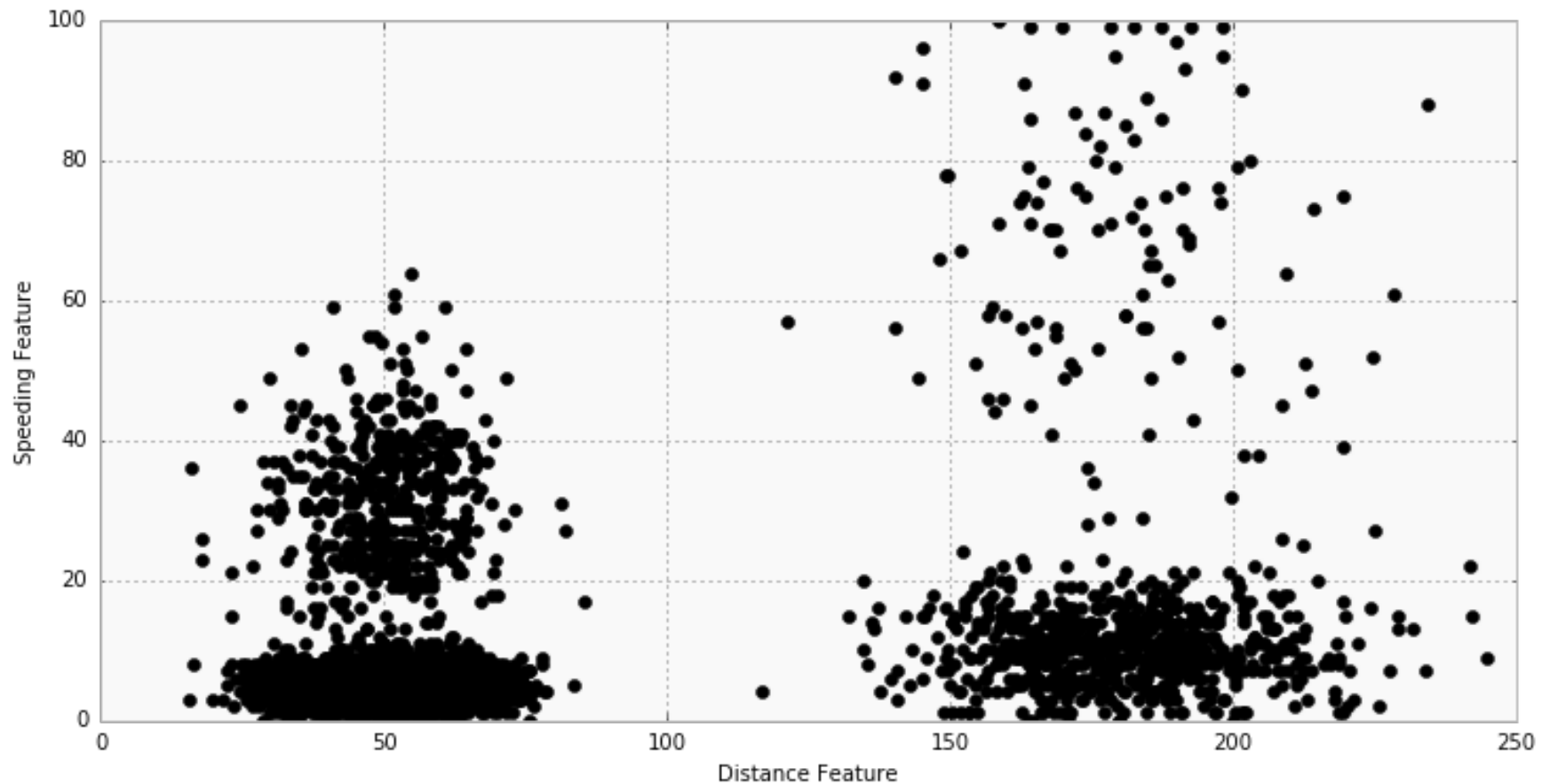
ML and DL – concepts and applications



ML and DL – concepts and applications

■ Unsupervised Learning

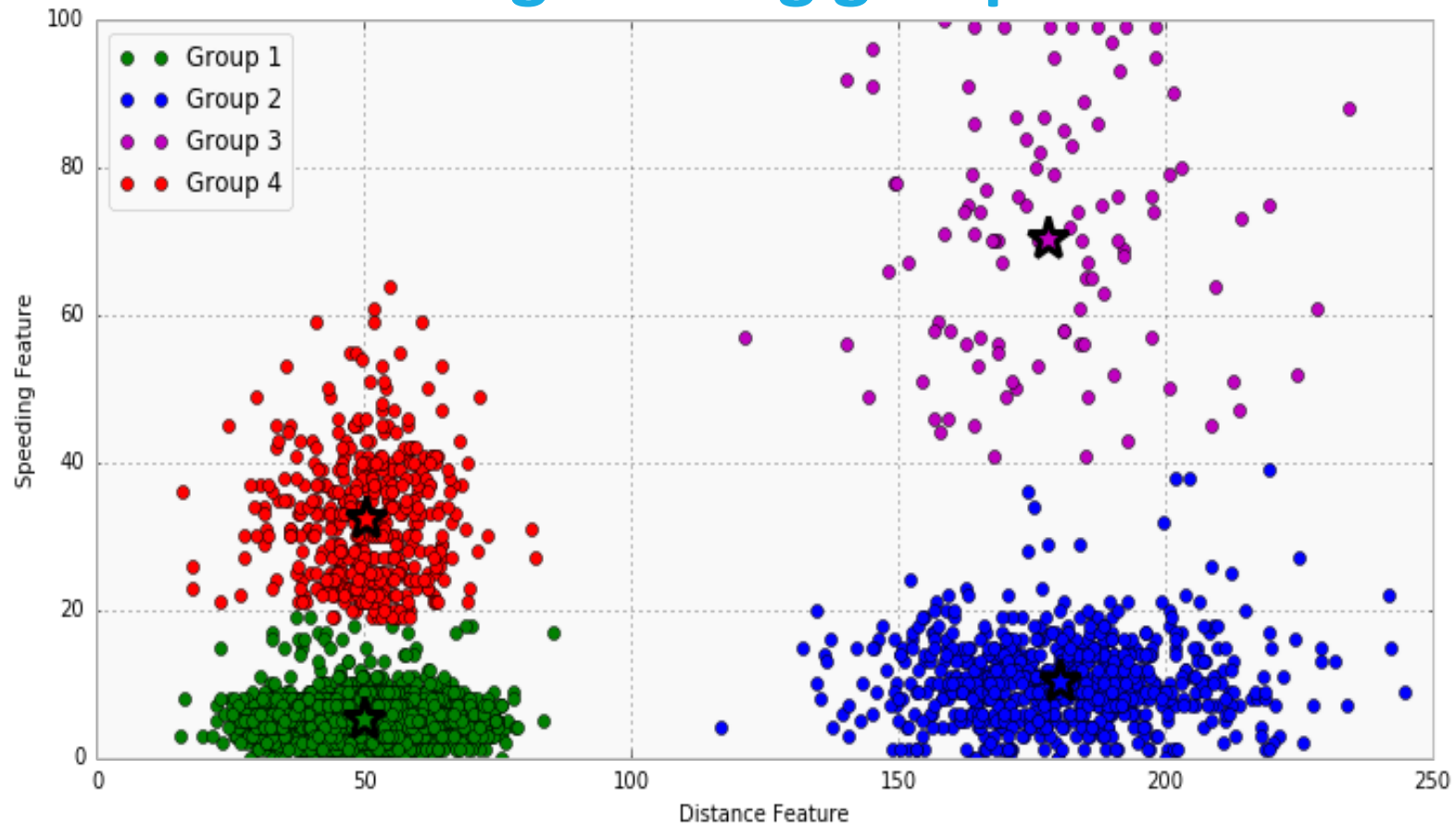
Clustering: finding groups in data



ML and DL – concepts and applications

■ Unsupervised Learning

Clustering: finding groups in data



ML and DL – concepts and applications

■ Unsupervised Learning

K-means clustering

Assume points are close to other points in group, far from points out of group

■ Algorithm:

1. Randomly initialize cluster centers



ML and DL – concepts and applications

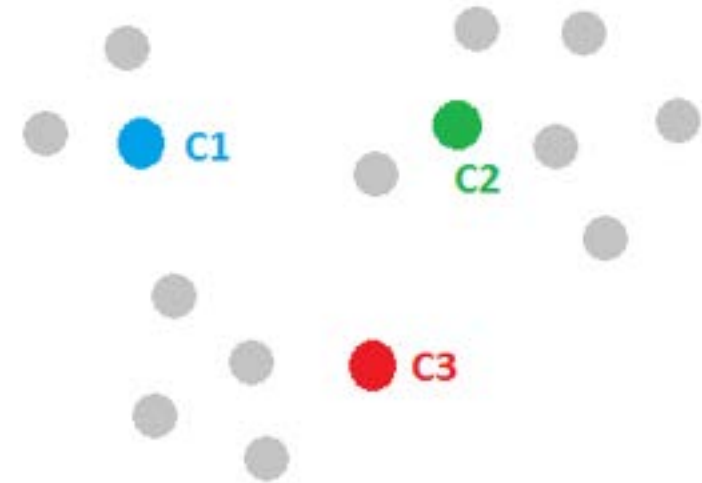
■ Unsupervised Learning

K-means clustering

Assume points are close to other points in group, far from points out of group

■ Algorithm:

1. Randomly initialize cluster centers



ML and DL – concepts and applications

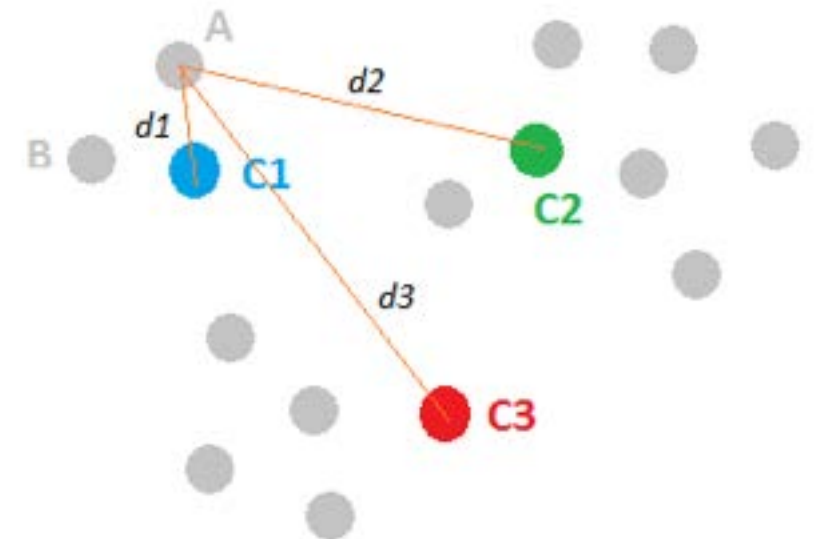
■ Unsupervised Learning

K-means clustering

Assume points are close to other points in group, far from points out of group

■ Algorithm:

1. Randomly initialize cluster centers
2. Calculate distance points $\leftrightarrow$ centers



ML and DL – concepts and applications

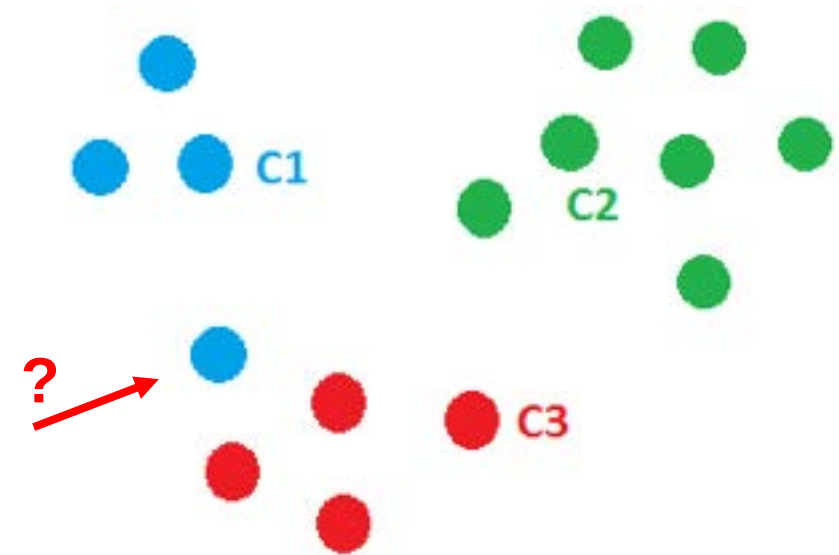
■ Unsupervised Learning

K-means clustering

Assume points are close to other points in group, far from points out of group

■ Algorithm:

1. Randomly initialize cluster centers
2. Calculate distance points $\leftrightarrow$ centers
3. Assign each point to closest cluster center



ML and DL – concepts and applications

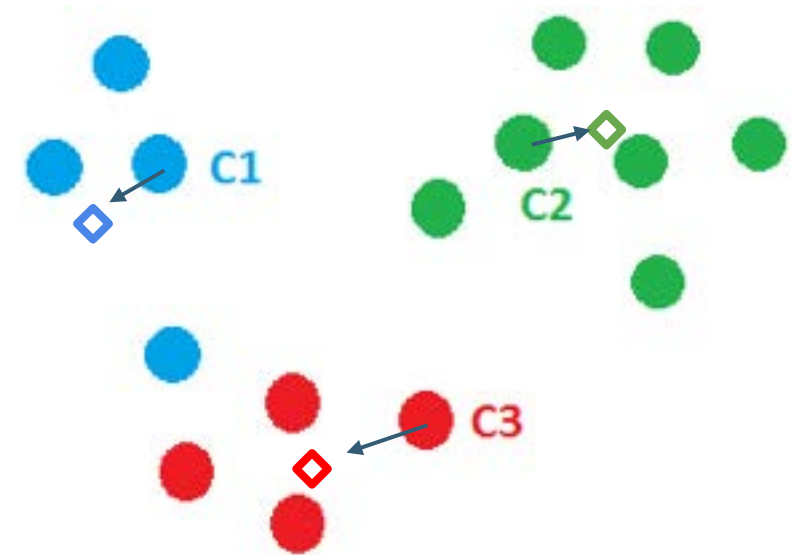
■ Unsupervised Learning

K-means clustering

Assume points are close to other points in group, far from points out of group

■ Algorithm:

1. Randomly initialize cluster centers
2. Calculate distance points $\leftrightarrow$ centers
3. Assign each point to closest cluster center
4. Update cluster centers: avg of points



ML and DL – concepts and applications

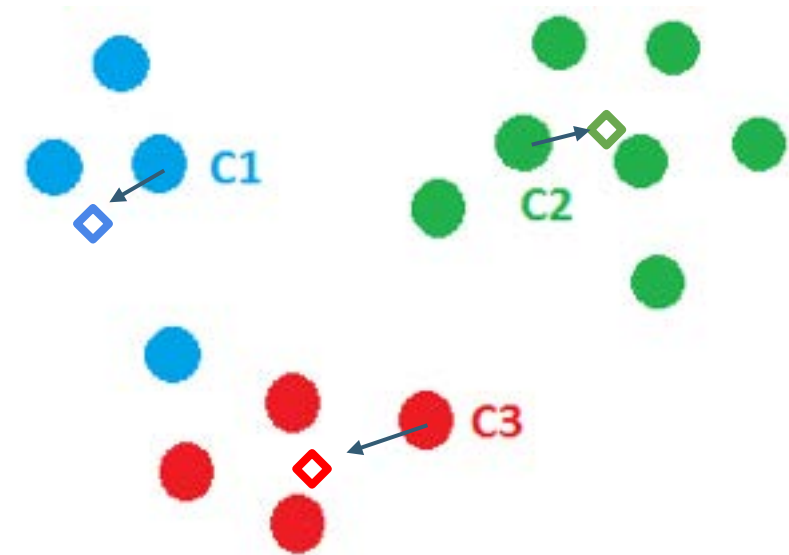
■ Unsupervised Learning

K-means clustering

Assume points are close to other points in group, far from points out of group

■ Algorithm:

1. Randomly initialize cluster centers
2. Calculate distance points $\leftrightarrow$ centers
3. Assign each point to closest cluster center
4. Update cluster centers: avg of points
5. Repeat!



ML and DL – concepts and applications

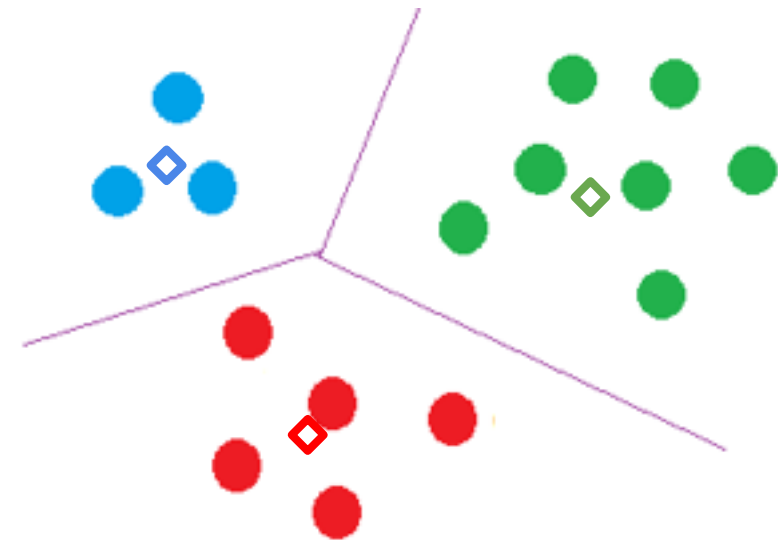
■ Unsupervised Learning

K-means clustering

Assume points are close to other points in group, far from points out of group

■ Algorithm:

1. Randomly initialize cluster centers
2. Loop until converged:
 - a) Calculate distance points $\leftrightarrow$ centers
 - b) Assign each point to closest center
 - c) Update cluster centers: avg of points



ML and DL – concepts and applications

■ Unsupervised Learning

K-means clustering – example: Clustering on images

Group together pixels by color, automatic segmentation: k-means, $k = 2$



ML and DL – concepts and applications

■ Unsupervised Learning

K-means clustering – example: Clustering on images

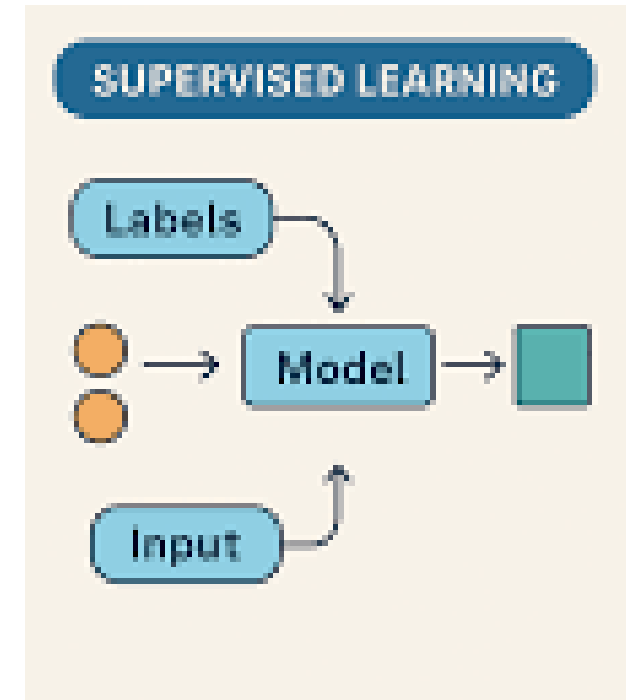
Group together pixels by color, automatic segmentation: k-means, $k = 4$



ML and DL – concepts and applications

■ Supervised Learning

- The model is trained using a dataset containing pairs of expected inputs and outputs.
- The goal is to teach the model how to link inputs to outputs.



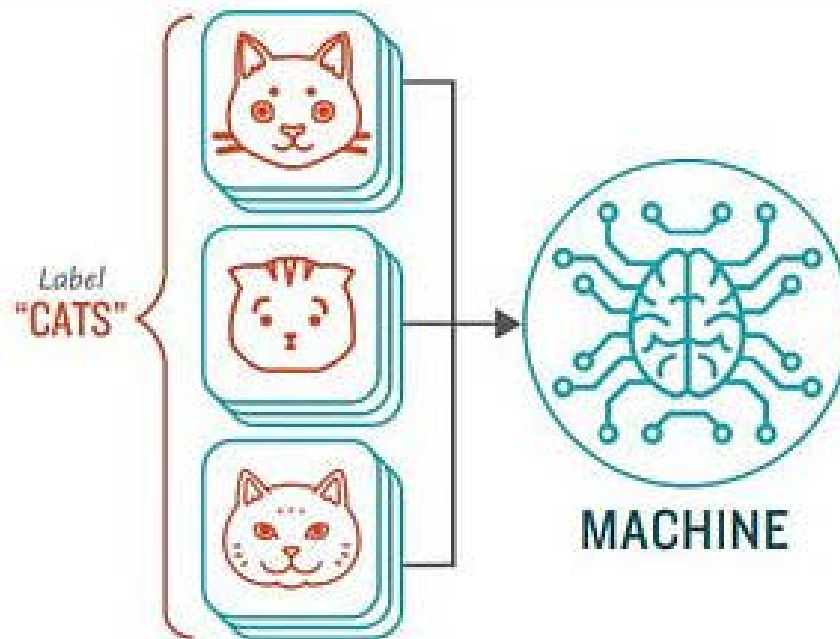
Given inputs to a function, try to predict the output
Have lots of labelled examples

ML and DL – concepts and applications

How **Supervised** Machine Learning Works

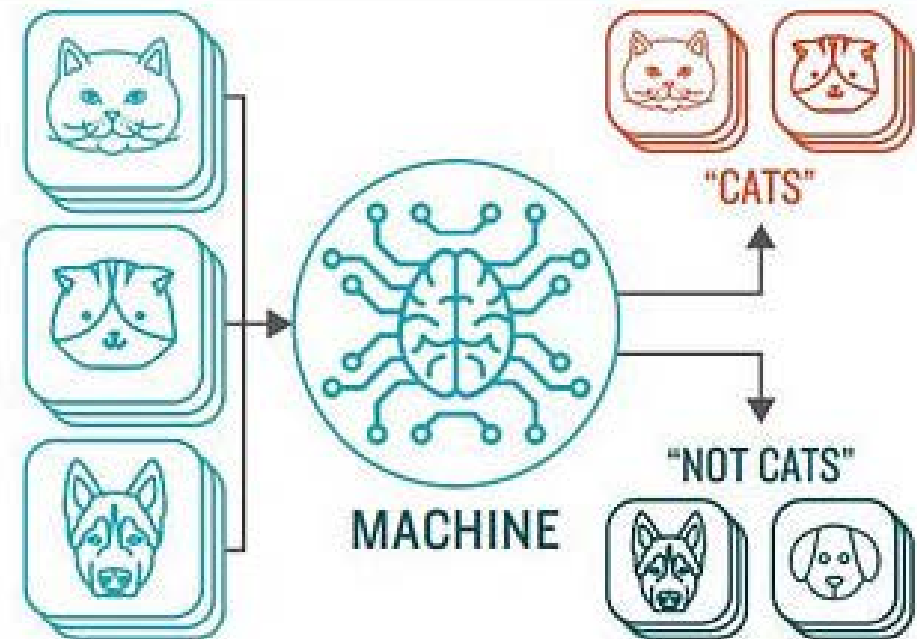
STEP 1

Provide the machine learning algorithm categorized or "labeled" input and output data from to learn



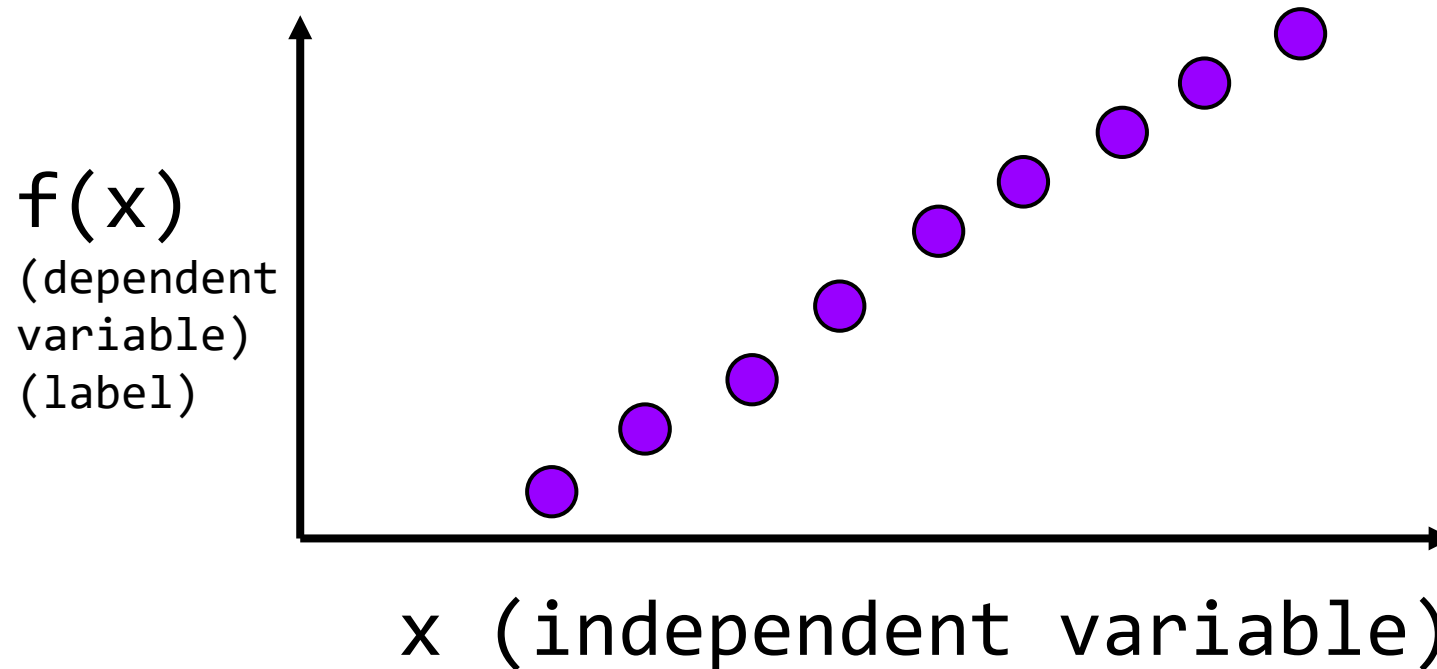
STEP 2

Feed the machine new, unlabeled information to see if it tags new data appropriately. If not, continue refining the algorithm



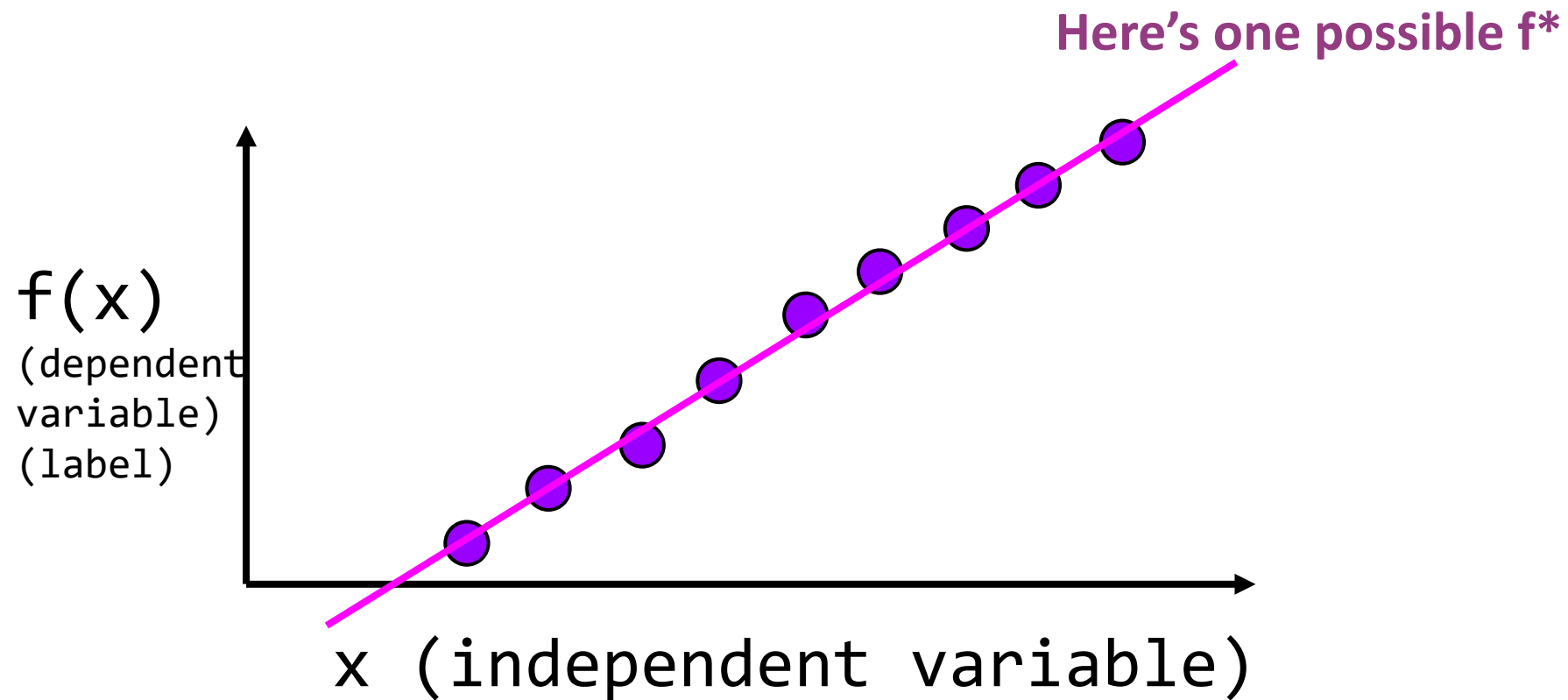
ML and DL – concepts and applications

- Supervised Learning: Want to estimate f



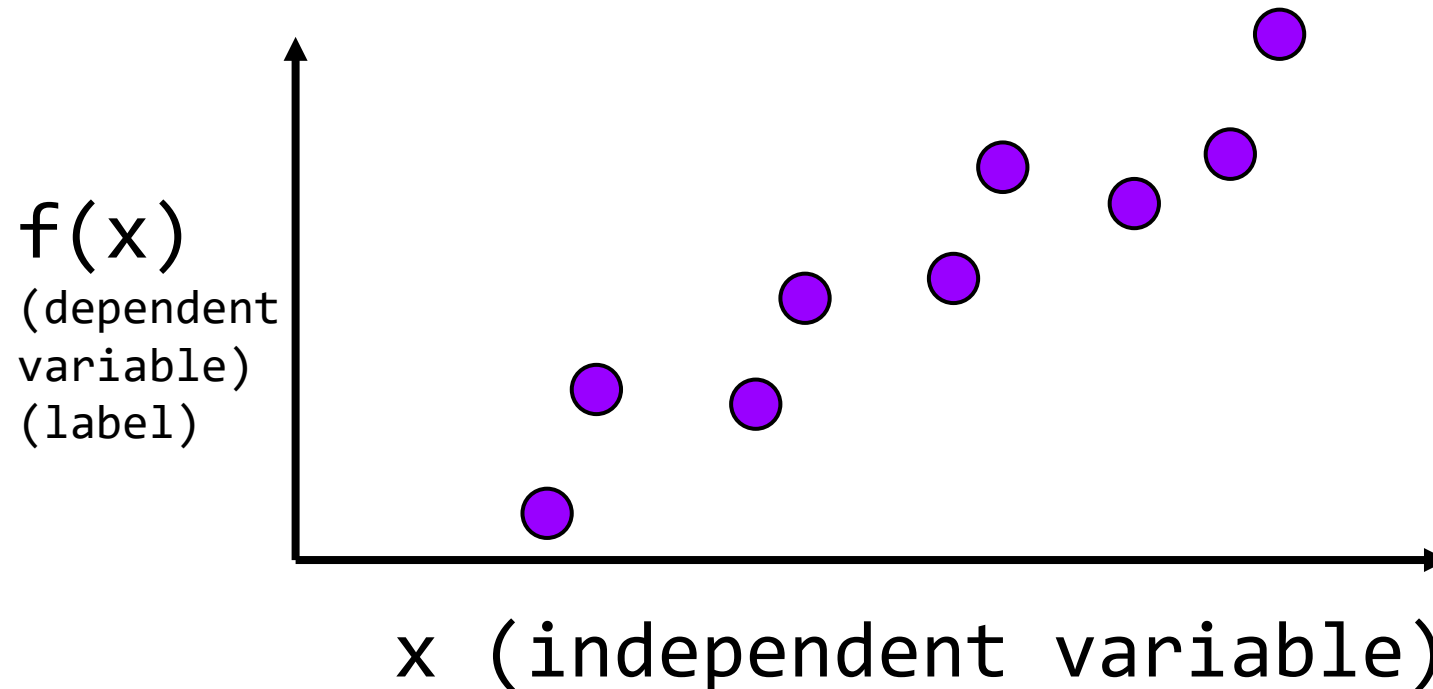
ML and DL – concepts and applications

- Supervised Learning: Want to estimate f



ML and DL – concepts and applications

- Supervised Learning: Data often has noise !



ML and DL – concepts and applications

■ Supervised Learning: Data often has noise !

Why?

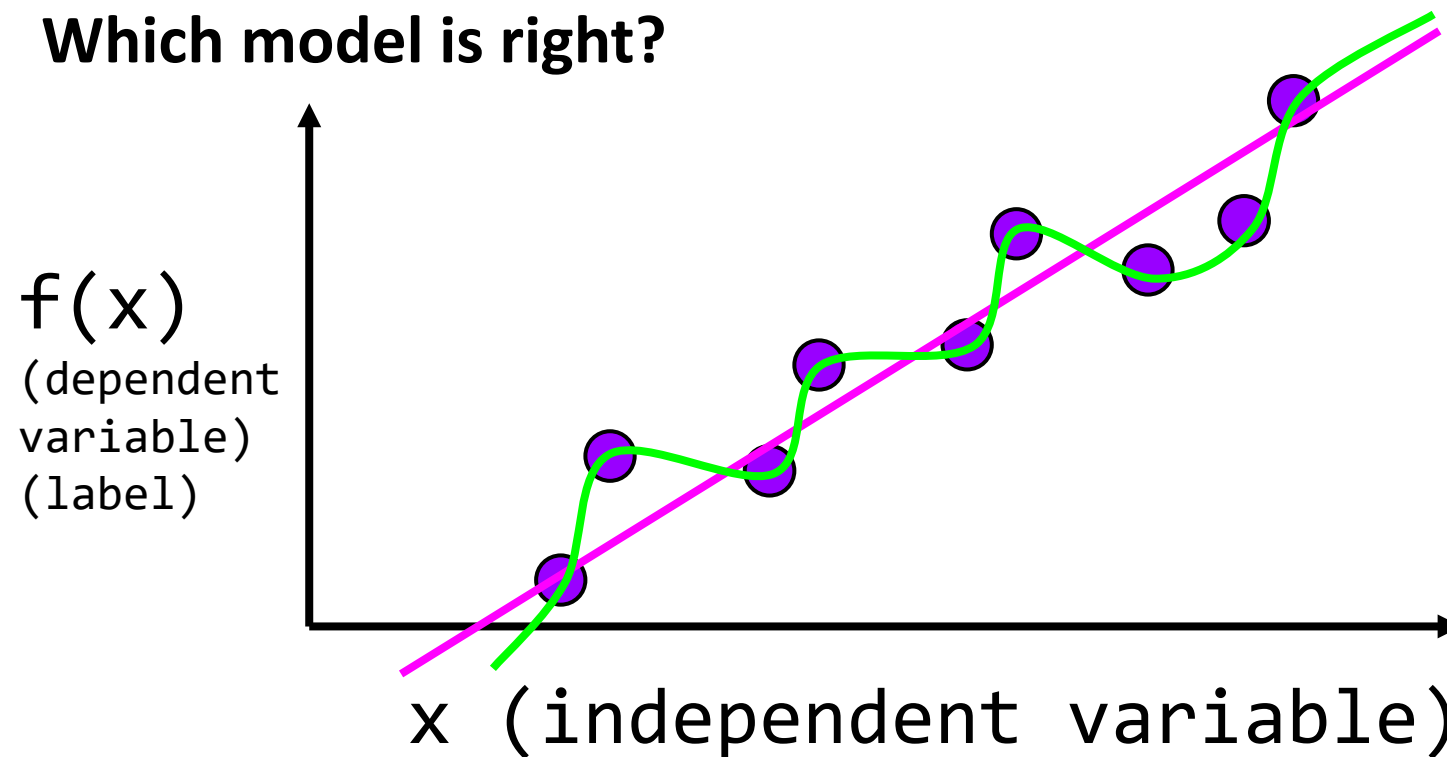
- **Randomness**
 - Static in phone lines, random distribution of photons hitting sensor, sensors aren't precise
- **Mislabelled data**
 - Common when humans label lots of data
- **Variables outside of model**
 - Variations might look like noise but are explained by a hidden, unknown variable

ML and DL – concepts and applications

■ Supervised Learning: Data often has noise !

Sometimes the data is more complex!

Which model is right?

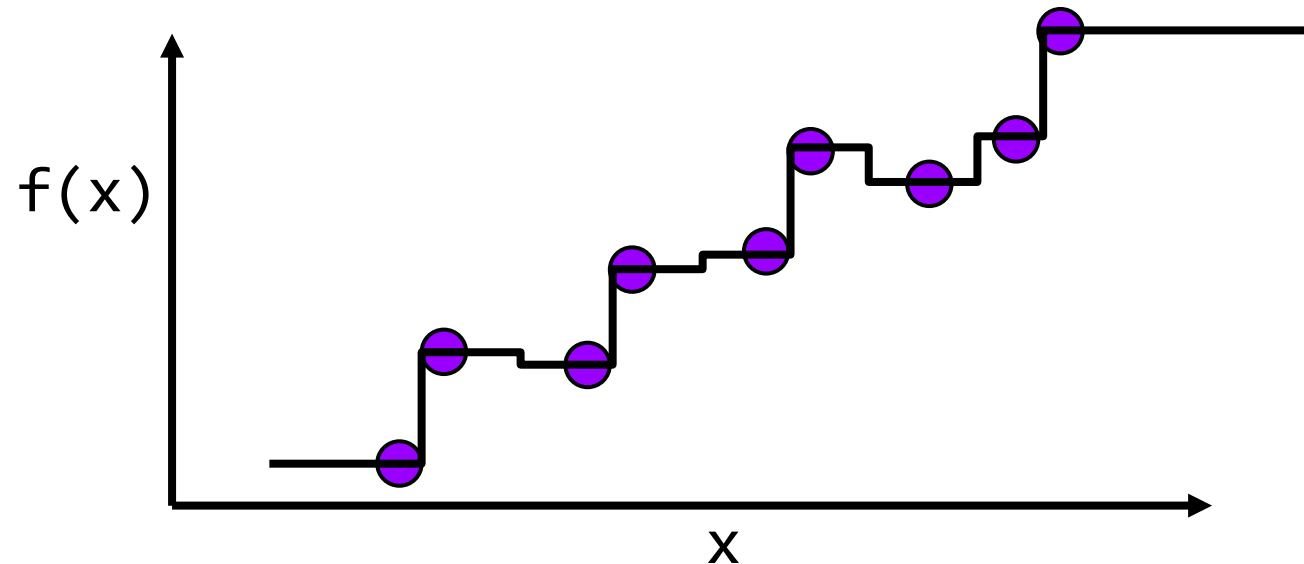


Noise
or
complex data?

ML and DL – concepts and applications

■ Supervised Learning: Nearest neighbor

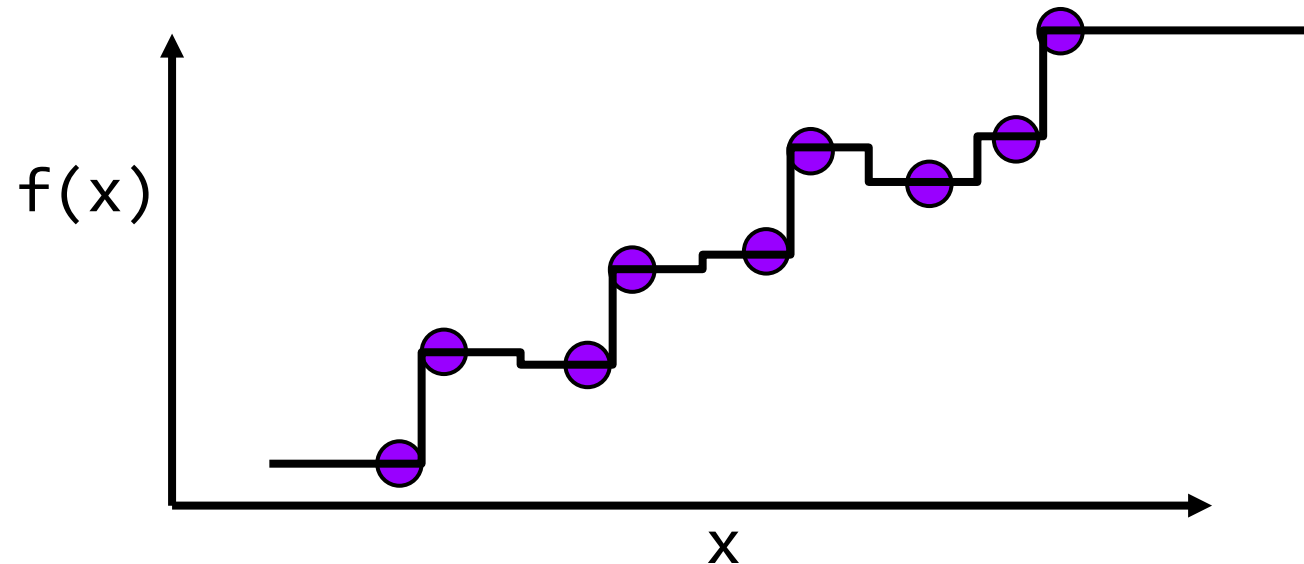
$f^*(x) = f(x')$ for nearest x' in training set



ML and DL – concepts and applications

■ Supervised Learning: Nearest neighbor

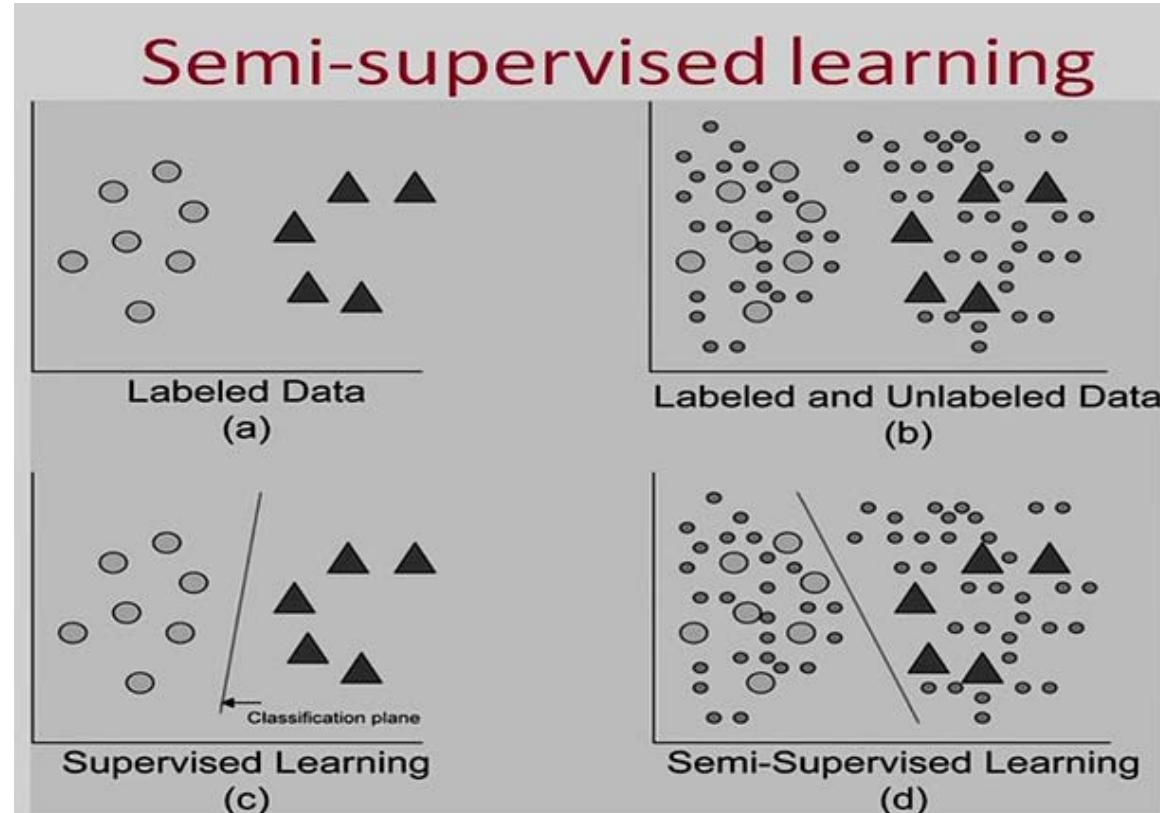
$f^*(x) = f(x')$ for nearest x' in training set



ML and DL – concepts and applications

■ Semi-Supervised Machine Learning

It combines elements of both supervisory and non-supervisory machine learning. The model is trained using a small set of data containing the predicted outputs and a larger set of unlabeled data.

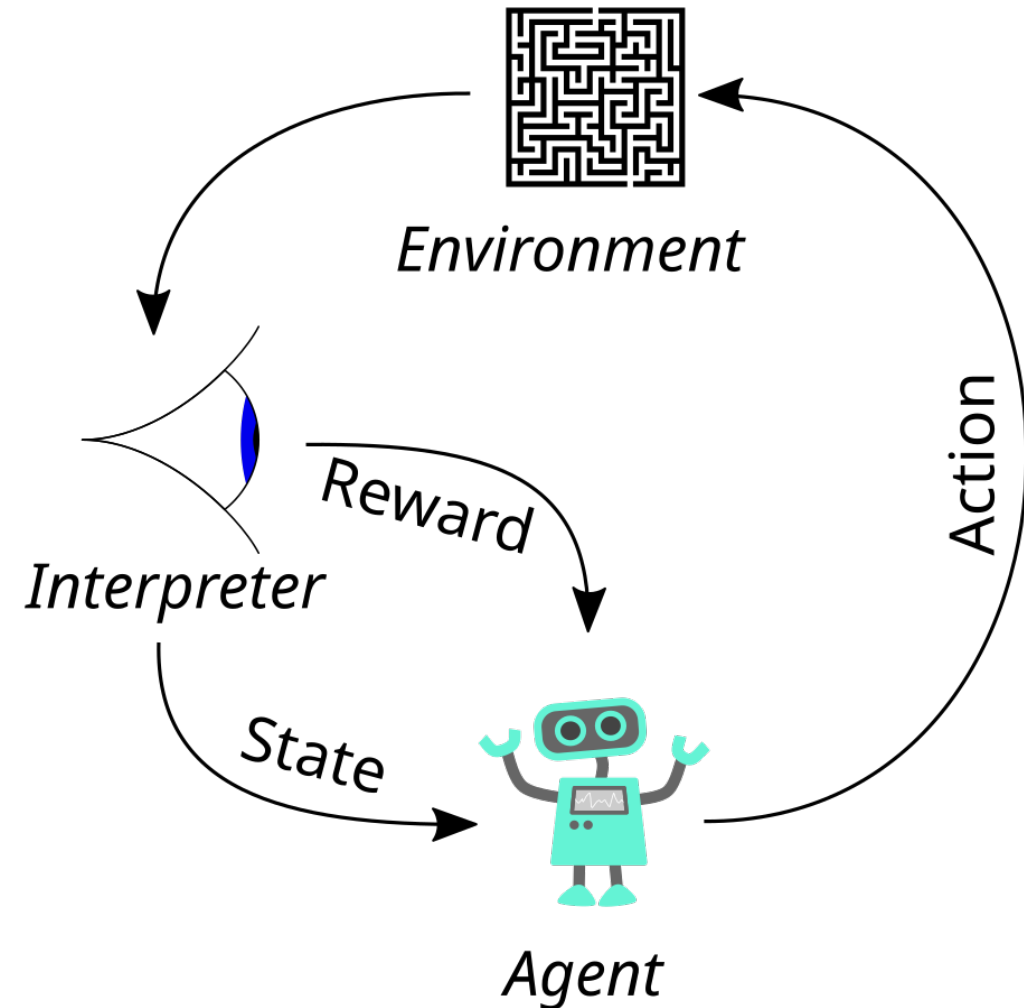


ML and DL – concepts and applications

■ Reinforcement Machine Learning

The model learns how to make decisions through interaction with its environment.

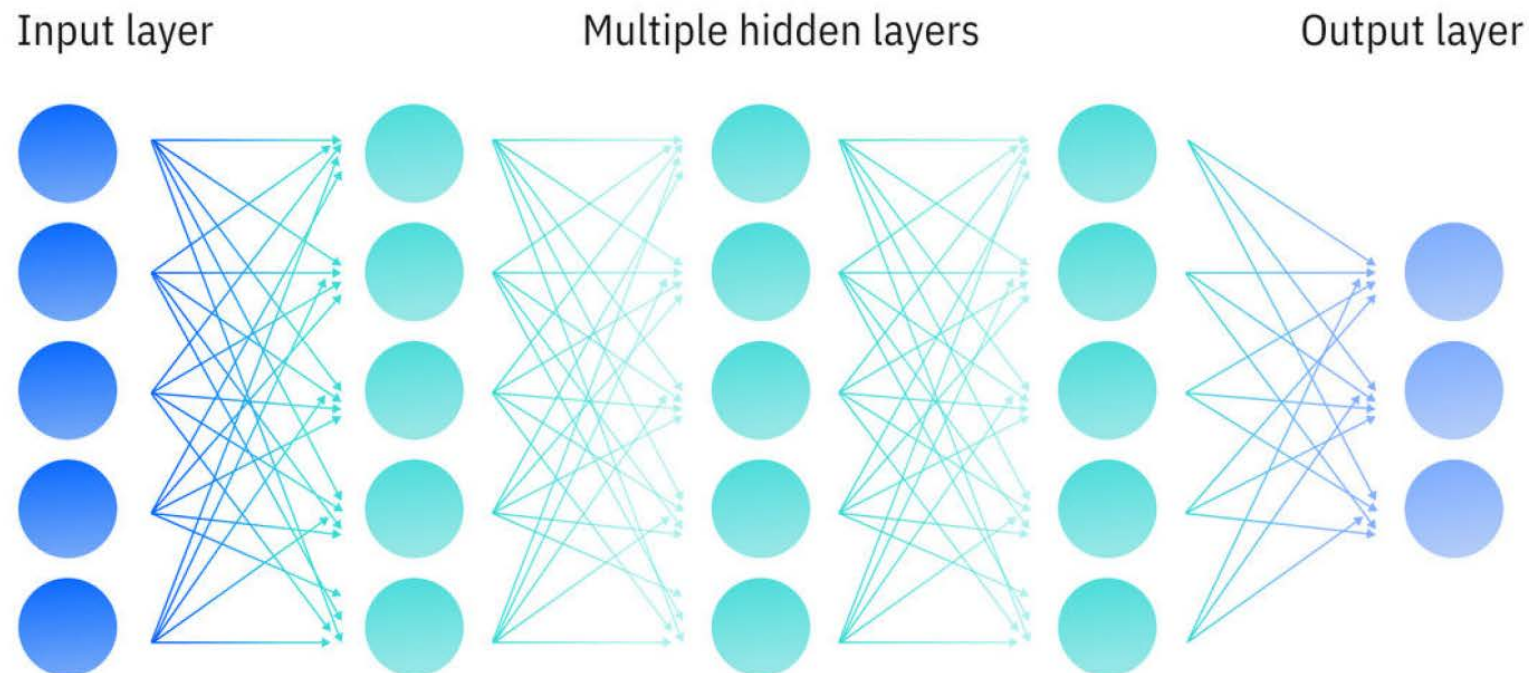
The model receives a reward or punishment based on its choices. The decisions he makes, which motivates him to make decisions that lead to achieving specific goals.



ML and DL – concepts and applications

■ Deep Learning

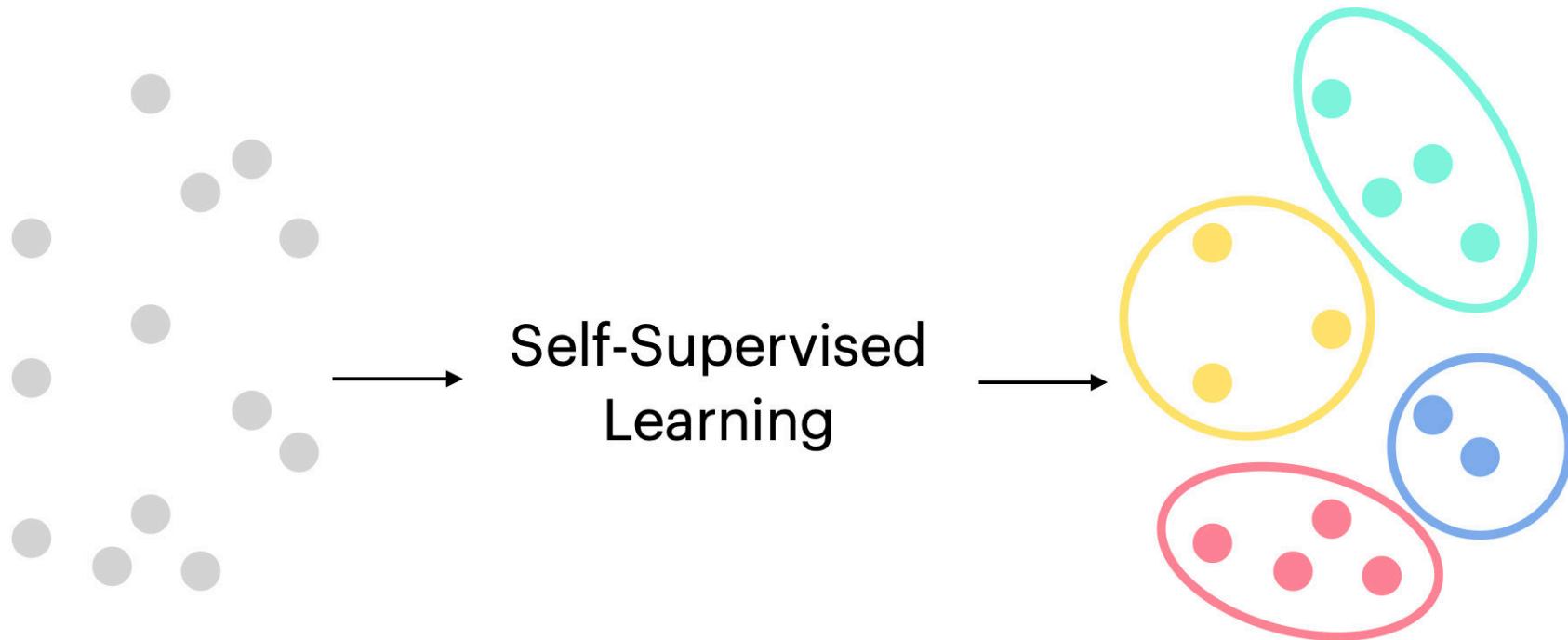
Deep neural networks are used to learn representations that encompass multiple levels of detail and concepts. They are widely used in fields such as computer vision and natural language processing.



ML and DL – concepts and applications

■ Self-Supervised Learning

- It is based on the idea of training the model using unlabeled data, where the expected outputs are generated from the data itself without the need for external labels.



ML and DL – concepts and applications

■ Transfer learning

The model learns from one task and then transfers the acquired knowledge to a related task, reducing the need for large amounts of data for each task.

